

Microeconomics — Practice Set with Solutions

100 questions covering every examinable concept • companion to the End-Term Notes

Written in the 2025 format: short descriptive questions, no multiple choice, answers judged on precision rather than length. Each Part gives the questions first and the worked solutions immediately after, so you can cover the solutions with a sheet of paper. Section references in square brackets point to the End-Term Notes. Every number here has been checked by computer.

How to use it. Do a Part at a time, closed book, writing full answers — the marks in this exam come from stating the condition and the reason, not from the final number alone. Then read the solution and check three things: did you name the right model, did you write the correct optimality condition, and did you sanity-check your answer.

Part	Questions	Coverage
A	1–14	Dominance, Nash, mixed strategies, games on discrete grids
B	15–27	Sequential games, bargaining, repeated games and collusion
C	28–33	Bayesian games, signalling, screening, adverse selection
D	34–45	Auctions, auctions with securities, contests and lotteries
E	46–54	Matching and market design, the core and the Shapley value
F	55–67	Risk and insurance, search, other solution concepts
G	68–90	Monopoly, oligopoly, price discrimination, tariffs, bundling, transfer pricing
H	91–100	Externalities, public goods, information, and the verbal questions

1 Part A — Dominance, Nash, mixing and discrete games

Questions

Q1. Consider the game below (row player's payoff first).

	<i>L</i>	<i>C</i>	<i>R</i>
<i>T</i>	4, 3	5, 1	6, 2
<i>M</i>	2, 1	8, 4	3, 6
<i>B</i>	3, 0	9, 6	2, 8

(a) Does either player have a strictly dominant strategy? (b) Solve the game by iterated deletion of strictly dominated strategies, stating what you delete at each round and why. (c) Find the Nash equilibrium. (d) Find each player's maximin strategy and explain why the maximin profile is not the equilibrium. [§1]

Q2. Explain, with a two-line example, why iterated deletion of *weakly* dominated strategies can give different answers depending on the order of deletion, while deletion of *strictly* dominated strategies cannot. [§1.1]

Q3. A row player has three strategies with payoffs $T = (10, 0)$, $M = (0, 10)$ and $B = (4, 4)$ against the column player's two strategies. Is B dominated? Would your answer change if B paid $(6, 6)$? [§1.1]

Q4. State whether each is true or false and justify in one sentence: (i) a strictly dominant strategy is played in every Nash equilibrium; (ii) a weakly dominant strategy is played in every Nash equilibrium; (iii) if every player has a strictly dominant strategy, the resulting profile maximises the sum of payoffs. [§1.1]

Q5. A worker chooses Shirk or Work; a manager chooses Inspect or Not, simultaneously. The worker gets 10 from shirking undetected, 0 from shirking and being caught, and 6 from working whatever happens. Inspection costs the manager 3; an undetected shirk costs her 8; a detected shirk costs her nothing beyond the inspection cost, and a working employee costs her nothing. (a) Find the mixed-strategy equilibrium. (b) The firm raises the penalty so that a caught shirker gets -5 instead of 0. What happens to the shirking rate and to the inspection rate? Explain the result. [§1.2]

Q6. In an asymmetric matching-pennies game, Row earns 9 at (T, L) and 1 at (B, R) and nothing otherwise, while Column earns 1 whenever the choices mismatch. Compute the mixed equilibrium and explain the "own-payoff" puzzle it produces. Which solution concept repairs it, and how? [§1.2, §14]

Q7. In Battle of the Sexes, payoffs are $(2, 1)$ if both choose Football, $(1, 2)$ if both choose Ballet, and $(0, 0)$ otherwise. Find the mixed equilibrium, each player's expected payoff, and the probability of miscoordination. Comment on how the mixed equilibrium compares with the pure ones. [§1.2]

Q8. Two firms sell an identical good, simultaneously post *integer* prices, supply whatever demand they get, and split demand equally at equal prices. Market demand is $Q = 12 - P$ and each firm's marginal cost is 3. (a) Find all pure-strategy Nash equilibria and the maximum equilibrium profit per firm. (b) What price would they set if they could collude? (c) Explain in one sentence why the answer to (a) differs from the textbook Bertrand result. [§2.1]

Q9. Four firms with commonly known values 3, 6, 9, 12 bid natural numbers for one object; ties are broken in favour of the higher valuation. (a) In the first-price sealed-bid auction, which firm must win, and what is the largest payoff any firm can earn in equilibrium? (b) In the second-price auction, what is the smallest bid with which the value-6 firm could win in equilibrium, and what is the largest payoff anyone can earn? [§2.2]

Q10. Two bidders compete for a prize of 200 in an all-pay auction; bids must be multiples of 10 and both bidders pay their bids. (a) What is the best reply to a rival bidding 100? (b) Show that there is no pure-strategy equilibrium. (c) What is each bidder's maximin strategy? [§2.3]

Q11. In the Traveler's Dilemma the reward and penalty are ± 2 . Suppose they are changed to ± 5 . Does the Nash equilibrium change? Does the maximin strategy change? What does experimental evidence say, and which behavioural model captures it? [§2.4, §14]

Q12. Modify the money-request game so that each player names an integer from 5 to 15, receives that amount, and earns a bonus of 12 for naming exactly one less than the other player. Find the maximin strategy and determine whether a pure-strategy Nash equilibrium exists. [§2.5]

Q13. Twenty consumers are located at the integer points $1, 2, \dots, 20$ on a line. Two shops simultaneously choose integer locations; each consumer buys from the nearer shop and splits evenly when equidistant. Find all pure-strategy Nash equilibria and name the political result this reproduces. [§2.6]

Q14. Two firms are locked in a war of attrition over a contract worth V_1 and V_2 respectively; each loses one unit of payoff per unit of time until one concedes. Is there an equilibrium with delay? Justify your answer, and say what the valuations do determine. [§2.6]

Solutions

S1. (a) **No.** T beats M at L ($4 > 2$) but loses at C ($5 < 8$), and Column's R beats C everywhere but loses to L at T ($2 < 3$).

(b) Round 1: for Column, R strictly dominates C — $2 > 1, 6 > 4, 8 > 6$ — so delete C . Round 2: in the remaining 3×2 game Row's payoffs are $T = (4, 6), M = (2, 3), B = (3, 2)$, so T strictly dominates both M and B ; delete them. Round 3: against T , Column compares 3 and 2 and picks L .

(c) (T, L) with payoffs $(4, 3)$. Verify directly in the original game: at L Row's best is T ($4 > 2, 3$); at T Column's best is L ($3 > 1, 2$).

(d) Row's worst payoffs are $T : 4, M : 2, B : 2$, so Row's maximin is T . Column's worst are $L : 0, C : 1, R : 2$, so Column's maximin is R . The profile (T, R) is not an equilibrium because maximin asks only "what is my worst case?" and ignores what the opponent actually wants; the two coincide only in zero-sum games.

S2. Strict domination is preserved by deletion: if s is strictly worse than s' against every opponent profile, it remains so in any subgame you are left with, so the final set does not depend on the order. Weak domination is not preserved, because the *tie* that made s only weakly worse may be created by an opponent strategy that a different deletion order removes first. Example: with two Row strategies that give identical payoffs against L but differ against R , deleting R first destroys the reason one of them was weakly dominated. The practical rule: **say which strategy you delete and why at each round**, and never claim uniqueness when deleting weakly dominated strategies.

S3. B is not dominated by T (it wins in the second column) nor by M (it wins in the first), but the mixture $\frac{1}{2}T + \frac{1}{2}M$ yields 5 against either column, and $5 > 4$, so B is **strictly dominated by a mixture** and is a never-best response. If B paid $(6, 6)$ then $6 > 5$ and no mixture of T and M can beat it; B would then be **undominated** and indeed the unique best reply to a belief near 50–50. The lesson: always check mixtures before declaring a strategy undominated.

S4. (i) **True** — any profile not using it admits a strictly profitable deviation. (ii) **False** — the opponent's actual play may make the tie irrelevant, which is why second-price auctions have equilibria in which nobody bids her value. (iii) **False** — the Prisoner's Dilemma has strictly dominant strategies and a Pareto-dominated outcome.

S5. (a) Write the manager's payoffs: Inspect costs 3 whatever the worker does, so it yields -3 ; Not yields -8 against a shirker and 0 against a worker. Let the worker shirk with probability p . The manager is indifferent when

$$-3 = p(-8) + (1 - p)(0) = -8p \Rightarrow p = \underline{3/8}.$$

Let the manager inspect with probability q . The worker is indifferent when $q(0) + (1 - q)(10) = 6$, i.e. $10 - 10q = 6$, so $q = \underline{0.4}$.

(b) The worker's indifference becomes $q(-5) + (1 - q)(10) = 6$, i.e. $10 - 15q = 6$, so $q = 4/15 \approx 0.267$. The *shirking* rate is unchanged at $3/8$, because it is pinned down by the *manager's* payoffs, which the penalty did not touch; what falls is the *inspection* rate. The economics: a harsher penalty lets the firm achieve the same deterrence with less costly monitoring — it saves on inspection rather than reducing shirking.

S6. Let $p = \Pr(T)$ and $q = \Pr(L)$. Column is indifferent when $p \cdot 0 + (1 - p) \cdot 1 = p \cdot 1 + (1 - p) \cdot 0$, i.e. $1 - p = p$, so $p = \underline{1/2}$ — *independent of Row's payoff of 9*. Row is indifferent when $9q = 1 - q$, so $q = \underline{1/10}$. The puzzle is that raising Row's own payoff changes Column's behaviour, not Row's, which contradicts experiments. **Quantal response equilibrium** repairs it: players choose better actions more often but not always, $\Pr(a) \propto e^{\lambda U(a)}$, so a larger payoff at (T, L) does raise $\Pr(T)$ — with $\lambda = 1$ it rises to about 0.89.

S7. Let the wife play Football with probability p . The husband is indifferent when $1 \cdot p = 2(1 - p)$, so $p = 2/3$; symmetrically the husband plays Football with probability $1/3$. Each then earns $\frac{2}{3} \cdot \frac{1}{3} \cdot 2 + \frac{1}{3} \cdot \frac{2}{3} \cdot 1 = 2/3$. Miscoordination occurs when both pick their own favourite or both pick the other's: $\frac{2}{3} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{3} = 5/9$. So the mixed equilibrium is worse for *both* players than either pure equilibrium — mixing buys symmetry at the cost of surplus, which is exactly the gap a correlation device (§14) closes.

S8. (a) Check (4, 4): each earns $(4 - 3)(12 - 4)/2 = 4$; undercutting to 3 earns $(3 - 3)(9) = 0$ and raising to 5 earns 0, so it is an equilibrium. (3, 3) is also an equilibrium, with zero profit. (5, 5) is *not*: each would earn $(5 - 3)(7)/2 = 7$, but undercutting to 4 earns $(4 - 3)(8) = 8 > 7$. So the maximum equilibrium profit is 4 each.

(b) Colluding, they maximise $(P - 3)(12 - P)$ over integers: $P = 7$ gives $4 \times 5 = 20$ and $P = 8$ gives $5 \times 4 = 20$, so **7 or 8**, worth 20 jointly.

(c) With continuous prices the smallest undercut is infinitesimal, so any price above cost unravels; with an integer grid the smallest undercut costs a whole rupee of margin, which can exceed the gain from doubling market share, so **a price one step above cost survives**.

S9. (a) For a bid b to win in equilibrium, the value-9 firm must not want to outbid it: $9 - (b + 1) \leq 0$, so $b \geq 8$. Only the value-12 firm can afford such a bid, so **the value-12 firm must win**, and its best equilibrium payoff is $12 - 8 = 4$ at a bid of 8.

(b) In the second-price auction the winner pays somebody else's bid, so a low-value firm can win by bidding high. For the value-6 firm to win, the value-12 firm must not want to outbid it and pay the value-6 firm's bid: $12 - b \leq 0$, so $b \geq 12$. The largest payoff anyone can earn is the value-12 firm bidding high while everyone else bids the minimum of 1, earning $12 - 1 = 11$.

S10. (a) **Bid 110**, netting $200 - 110 = 90$; matching at 100 wins only half the time for an expected $100 - 100 = 0$, and bidding less loses the money outright.

(b) In any profile, the loser is paying a positive bid for nothing and would rather bid 0; and if the loser bids 0, the winner would rather drop to 10. Neither can be an equilibrium, so **no pure equilibrium exists**.

(c) **Bid 0**: it guarantees a payoff of 0 (or 100 if the rival also bids 0), whereas every positive bid can be lost outright.

S11. The Nash equilibrium does not change: it remains the lowest claim, 2. The iterated argument only needs the reward to exceed 1, so that undercutting by one unit always pays; a larger penalty makes the unravelling *faster* but not different. **The maximin strategy also stays at 2**, since claiming 2 guarantees 2 while any higher claim can pay $2 - 5 = -3$ against a rival claiming 2 — in fact the higher penalty makes maximin more attractive. Experimentally, claims fall sharply as the penalty rises even though theory says nothing changes; **level- k** reasoning captures this, since a larger penalty makes each level of reasoning cut more aggressively.

S12. Maximin is 15: it guarantees 15, while any lower request x guarantees only $x < 15$. **There is no pure-strategy Nash equilibrium** (verified by checking all 11×11 profiles). The argument: at any profile where the two requests differ by more than one, the lower player gains by moving to one below the higher; at a matched pair (k, k) with $k \geq 6$, moving to $k - 1$ earns $k - 1 + 12 > k$; and at (5, 5) moving up to 6 earns $6 > 5$. So every profile has a profitable deviation.

S13. Both shops locate at the **median**. With 20 consumers the median lies between positions 10 and 11, so the pure equilibria are (10, 10), (10, 11), (11, 10) and (11, 11). At any other profile the shop further from the centre gains by moving one step towards its rival. This is the **median voter theorem**: with single-peaked preferences on a line and majority rule, the median voter's ideal point beats every alternative, which is why two-party platforms converge.

S14. No: in every Nash equilibrium one firm concedes immediately. Suppose both planned to hold out for a positive length of time. Whichever firm would eventually concede is burning payoff during the wait and then getting nothing, so it does strictly better by conceding at time zero and saving the waiting cost; and if it would *never* concede, the other firm's plan to wait is not a best response either. What the valuations determine is **which** firm concedes across the different equilibria — there is an equilibrium in which either one gives way — but never *whether* there is delay. (Delay reappears only in the incomplete-information version, where each firm is unsure of the other's value.)

2 Part B — Sequential games, bargaining and repetition

Questions

Q15. An incumbent monopolist earns 120. An entrant chooses Enter or Stay out. If it enters, the incumbent chooses Accommodate, giving each 45, or Fight, giving each -30 . If the entrant stays out it earns 0 and the incumbent keeps 120. (a) Find the Nash equilibria and the subgame-perfect equilibrium. (b) Suppose the incumbent can first sink an investment of 20 in capacity that would raise its fighting payoff from -30 to $+10$ while leaving its accommodation payoff at $45 - 20 = 25$. Should it invest, and why does this change the entrant's decision? [§3.1]

Q16. In the King of the Hill game the payoffs are: King at the end 10, Subject at the end 6, Dethroned King 2. (a) With N subjects moving one at a time, is the incumbent dethroned? (b) Now suppose a Dethroned King earns 7 instead of 2. How does the answer change, and why? [§3.2]

Q17. A pile has 10 coins. Players alternate removing 1, 2 or 3 coins, and whoever takes the last coin wins. (a) Which piles are losing for the player to move? (b) What is the winning first move from 10? (c) State the general invariant.

[§3.3]

Q18. Four pirates, ranked 1 (most senior) to 4, divide 100 coins. The most senior proposes a split; all pirates vote, including the proposer; the proposal passes on a majority, otherwise the proposer is thrown overboard and the next most senior proposes. Pirates prefer coins, then survival takes priority over coins. Solve by backward induction and state the general lesson. [§3.4]

Q19. Three options A, B, C and three voters with preferences $A \succ B \succ C$, $B \succ C \succ A$ and $C \succ A \succ B$. (a) Show that majority preference cycles. (b) You control the agenda of two successive pairwise votes and want A to win. What order do you set? (c) What does this imply about “the will of the majority”? [§3.4]

Q20. A proposes a split of 600. If B rejects, B proposes a split of 400. If A rejects that, A proposes a split of 200, and if B rejects the last proposal both get 0. Indifference means acceptance. Find the subgame-perfect outcome and explain why no delay occurs on the equilibrium path. [§4.1]

Q21. In the infinite-horizon alternating-offer game, player 1 proposes first and the discount factors are $\delta_1 = 0.8$ and $\delta_2 = 0.9$. (a) Find the equilibrium shares. (b) Who does better, and is that surprising given who moves first? (c) What happens as both discount factors approach 1? [§4.2]

Q22. A firm and a worker can create a surplus of 200. If they disagree the firm earns 60 and the worker earns 40. (a) Find the Nash bargaining split. (b) The worker acquires a certificate that raises her outside option to 100. What is the new split, and how much is the certificate worth to her? (c) Explain why the answer to (b) is not the full increase in her outside option. [§4.3]

Q23. A supplier must sink 50 into a machine that is worth 200 only to one particular buyer and nothing to anyone else. After the investment the two bargain and split the surplus equally. (a) Does the supplier invest? (b) Name the problem and give two contractual fixes. [§4.3]

Q24. In a repeated Prisoner’s Dilemma the per-period payoffs are: mutual cooperation 8, one-off defection 12, mutual defection 3, and the sucker’s payoff 1. Players use grim trigger. (a) Find the minimum discount factor that sustains cooperation. (b) Interpret the formula in one sentence. [§5.1]

Q25. Four identical firms compete in prices with a homogeneous product and constant marginal cost, and consider colluding at the monopoly price. (a) Derive the minimum discount factor. (b) What happens as the number of firms grows, and why? (c) Why is the threshold *lower* in a Cournot industry with the same number of firms? [§5.3]

Q26. Take the cooperation payoffs $R = 10$, $T = 16$, $P = 4$. (a) Find the grim-trigger threshold when defection is detected immediately. (b) Now suppose a defection is only detected after two periods, so the defector enjoys T for two periods before punishment begins. Find the new threshold. (c) Comment. [§5.3]

Q27. Explain in about 60 words why price-match guarantees — which sound like aggressive competition — make collusion easier among firms selling near-identical products. [§5.5]

Solutions

S15. (a) Nash equilibria: (Enter, Accommodate) with (45, 45), and (Stay out, Fight) with (0, 120) — the second survives as a Nash equilibrium only because the threat is never tested. Backward induction: once entry has occurred the incumbent compares 45 with -30 and accommodates, so the entrant enters. The SPE is **(Enter, Accommodate)**; the fighting equilibrium relies on an **incredible threat**.

(b) After the investment, fighting pays $+10$ against accommodating 25, so fighting is still not a best response — the investment as described does *not* make the threat credible, and the incumbent earns 25 rather than 45. **It should not invest** on these numbers. For commitment to work, the investment must make fighting *strictly better than accommodating after entry*; then the entrant, anticipating a fight, stays out and the incumbent keeps $120 - 20 = 100$. The general point: **commitment works by destroying your own option to be soft**, and it must be irreversible and observable.

S16. (a) **Dethroned if and only if N is odd**, exactly as in the original game. The ranking that drives the argument is King (10) > Subject (6) > Dethroned (2), so the last subject always charges (10 > 6), and subject k charges exactly when an even number of subjects follow him. The specific numbers do not matter, only the ordering.

(b) With Dethroned = 7 > Subject = 6, being dethroned is better than never having been king. Now charging is a **dominant** action for every subject at every point — the worst outcome from charging (7) beats the best from idling (6) — so **everybody always charges and the incumbent is always dethroned, whatever the parity of N** . The parity result depends entirely on the dethroned king being the *worst* role.

S17. (a) The player to move **loses exactly when the pile is a multiple of 4**: from 0, 4, 8, ... every legal move (1, 2 or 3) hands the opponent a non-multiple, and from a non-multiple you can always return to a multiple.

(b) $10 = 4(2) + 2$, so **take 2 coins and leave 8**.

(c) With a maximum take of k , the losing positions are the **multiples of $k + 1$** ; the winning move is to leave the opponent at the next lower multiple. State the invariant — that is where the marks are, not the arithmetic.

S18. Work upwards. *Two pirates (3 and 4 left)*: pirate 3 proposes, votes for himself, and a tie passes, so he takes all 100 and pirate 4 gets 0. *Three pirates (2, 3, 4)*: pirate 2 needs one more vote. Pirate 4 would get 0 if pirate 2 is thrown overboard, so one coin buys his vote: **(99, 0, 1)** for pirates 2, 3, 4. *Four pirates*: pirate 1 needs two votes including his own, so he needs one more. Pirate 3 would get 0 in the continuation, so one coin buys him: **(99, 0, 1, 0)**. The lesson:

proposal power combined with a known continuation is worth almost everything, because each pirate only has to be paid one coin more than his continuation value.

S19. (a) A beats B (voters 1 and 3), B beats C (voters 1 and 2), C beats A (voters 2 and 3): the majority relation **cycles**, so there is no Condorcet winner.

(b) Vote B **against** C **first** — B wins — **then** B **against** A , which A wins.

(c) With cyclic majority preferences, “the will of the majority” is not well defined, and **whoever controls the agenda controls the outcome**. This is why procedural rules are contested, and it is the same impossibility that Arrow’s theorem generalises.

S20. Backward induction, round by round. In the last round A proposes a split of 200 and B accepts anything non-negative, so A takes 200 and B gets 0. In the middle round B proposes a split of 400 knowing A would get 200 by rejecting, so B offers A exactly 200 and keeps 200. In the first round A proposes a split of 600 knowing B would get 200 by rejecting, so A offers B 200 and keeps **400**. Outcome: **A gets 400, B gets 200, accepted immediately**.

No delay occurs because both players can compute the continuation: any offer that gives the responder her continuation value is accepted, and the proposer never has a reason to offer more or to force a round of shrinkage. **The threat of delay shapes the terms; it is never carried out.**

S21. (a) *Where the formula comes from.* Let X be player 1’s share when she proposes and Y her share when he proposes. A proposer gives the responder exactly her discounted continuation value, so $X = 1 - \delta_2(1 - Y)$ and $Y = \delta_1 X$. Substituting the second into the first gives $X(1 - \delta_1\delta_2) = 1 - \delta_2$, hence

$$X = \frac{1 - \delta_2}{1 - \delta_1\delta_2} = \frac{1 - 0.9}{1 - 0.72} = \frac{0.1}{0.28} = \underline{0.357},$$

so player 1 gets about 35.7% and player 2 about **64.3%**.

(b) **Player 2 does better despite moving second**, because he is more patient ($\delta_2 > \delta_1$). Patience beats the first-mover advantage here: player 1 knows that delay costs her 20% per round and him only 10%, so she must concede.

(c) As $\delta_1, \delta_2 \rightarrow 1$ the split tends to **50–50** and the first-mover advantage vanishes, since waiting one round becomes costless. This limit is the symmetric Nash bargaining solution — the “Nash programme” link between the non-cooperative and axiomatic approaches.

S22. (a) Surplus above disagreement is $200 - 60 - 40 = 100$, split equally: firm $60 + 50 = \underline{110}$, worker $40 + 50 = \underline{90}$.

(b) New surplus above disagreement is $200 - 60 - 100 = 40$: firm $60 + 20 = \underline{80}$, worker $100 + 20 = \underline{120}$. The certificate is worth $120 - 90 = \underline{30}$ to her.

(c) Her outside option rose by 60 but she gained only 30, because **half of any improvement in your threat point leaks to the other side**: formally $\partial u_1 / \partial d_1 = \frac{1}{2}$ in the symmetric Nash solution. The same arithmetic explains why a firm gains from cultivating a second supplier even if it never uses one.

S23. (a) If he invests, the surplus of 200 is split equally, so he receives 100 against an investment of 50 — a net 50, so on these numbers he *does* invest. But note what he is not paid for: he creates 200 and captures 100, so if the investment cost were anywhere between 100 and 200 he would **refuse to invest even though the investment is socially worthwhile**. That is the failure.

(b) The problem is **hold-up**: relationship-specific investment plus ex-post bargaining means the investor anticipates being expropriated of half the return, so she under-invests. Fixes: a **long-term contract** agreed *before* the investment that fixes the price; **vertical integration** so that the two parties are one firm; or giving the investing party the bargaining power (for instance through an option or a break fee).

S24. (a) Compare the two infinite streams, using $\sum \delta^t = \frac{1}{1-\delta}$: cooperating forever pays $\frac{R}{1-\delta}$, while defecting once and then being punished forever pays $T + \frac{\delta P}{1-\delta}$. Multiplying through by $(1 - \delta) > 0$ gives $R \geq T(1 - \delta) + \delta P$, i.e. $R - T \geq \delta(P - T)$, and dividing by the *negative* number $P - T$ flips the inequality:

$$\delta \geq \frac{T - R}{T - P} = \frac{12 - 8}{12 - 3} = \frac{4}{9} \approx \underline{0.444}.$$

(b) The numerator is the *one-period* temptation from defecting, and the denominator is the *per-period* loss from being punished forever; cooperation survives when the future is weighted heavily enough that a permanent loss of $R - P$ outweighs a single gain of $T - R$. Note the sucker’s payoff of 1 never enters, because grim trigger never puts you in that position on the equilibrium path.

S25. (a) Colluding at the monopoly price gives each firm π^m/n ; defecting by an ε undercuts captures the whole market for one period, giving π^m ; the punishment is competitive pricing with zero profit. So

$$\delta \geq \frac{\pi^m - \pi^m/n}{\pi^m - 0} = 1 - \frac{1}{n}, \quad \text{with } n = 4 : \delta \geq \underline{0.75}.$$

(b) The threshold **risks towards 1**: with more firms your collusive share shrinks while the one-period prize from undercutting stays the whole market, so the temptation grows and collusion becomes harder. This is the formal version of “cartels with many members fall apart.”

(c) Because the Cournot punishment payoff is **positive** rather than zero, the *threat* is weaker — but so is the temptation,

since a Cournot deviator cannot capture the entire market, only expand output. The second effect dominates, and the linear Cournot duopoly threshold is $9/17 \approx 0.53$ against 0.5 for Bertrand. State the trade-off rather than just the number.

S26. (a) $\delta \geq \frac{16-10}{16-4} = 0.5$.

(b) With a two-period lag the defector earns T for two periods before punishment starts:

$$\frac{R}{1-\delta} \geq T(1+\delta) + \frac{\delta^2 P}{1-\delta}.$$

Multiplying by $(1-\delta)$: $10 \geq 16(1-\delta^2) + 4\delta^2 = 16 - 12\delta^2$, so $12\delta^2 \geq 6$ and $\delta \geq \sqrt{0.5} = 0.707$.

(c) Slow detection makes collusion much harder — the required patience jumps from 0.5 to 0.71. This is why real cartels invest so heavily in monitoring, information exchange and public list prices, and why secret discounts are so corrosive.

S27. *Model answer.* “A price-match guarantee removes the gain from undercutting. If I cut my price, my rival matches it immediately, so I capture no extra customers and simply lower the price for everybody — today and in every future period. The one-period temptation collapses towards the collusive payoff, so the grim-trigger condition holds at far lower discount factors. The guarantee therefore functions as an automatic, perfectly credible punishment mechanism, which is exactly what tacit collusion needs.”

3 Part C — Bayesian games, signalling and adverse selection

Questions

Q28. Player 1 has two types, y and n , equally likely and privately known; player 2 has no private information. Both choose B or S . Explain what a strategy is for each player, how many strategies each has, and state precisely what must be checked for a profile to be a Bayes–Nash equilibrium. Why is it wrong to average over your *own* type? [§6.1]

Q29. Two firms compete in quantities with inverse demand $P = 100 - Q$. Firm 1’s marginal cost is 20 and is common knowledge. Firm 2’s marginal cost is 10 or 30 with equal probability, known only to firm 2. Find the Bayes–Nash equilibrium quantities. [§6.2]

Q30. A firm pays 4,000 to high-productivity workers and 1,500 to low-productivity workers, and requires education level y for the high-wage contract. Education costs 250 per unit for low types and 100 per unit for high types. (a) Find the range of thresholds y^* that separates the types. (b) What is the minimum such threshold, and which constraint binds? (c) If the low type’s cost rose to 500 per unit, what happens to the minimum threshold, and what does that tell you about the social cost of signalling? [§6.3]

Q31. A buyer may bid B for a firm whose value v to its current owner is uniform on $[5, 25]$; the owner knows v and accepts iff $B \geq v$. In the buyer’s hands the firm is worth $1.4v$. (a) Find the optimal bid and the expected profit. (b) What would the optimal bid be if v were uniform on $[0, 25]$, and why does the answer change so dramatically? [§6.5]

Q32. Explain the difference between signalling and screening, giving one example of each, and state the condition without which neither works. [§6.3–6.4]

Q33. A used-car market has sellers whose values are uniform on $[0, 40]$; a car is worth $1.5v$ to a buyer. A certification body offers, for a fee of 10, a certificate that is awarded to any seller who pays — it verifies nothing. Will a separate, higher-priced certified market emerge? Show your reasoning. [§6.5]

Solutions

S28. A strategy for player 1 is a **plan with one action per type**, written (action if y , action if n), so with two types and two actions she has $2^2 = 4$ strategies: (B, B) , (B, S) , (S, B) , (S, S) . Player 2 has one information set and therefore 2 strategies.

A profile is a Bayes–Nash equilibrium if **every type of every player is best responding**: for each type of player 1 separately, the action prescribed must maximise her expected payoff given player 2’s strategy; and player 2’s action must maximise his expected payoff, averaging over player 1’s types using the prior and over player 1’s type-contingent actions.

Averaging over your *own* type is wrong because **you know your own type when you act**. The prior matters only for the *uninformed* party. A very common error is to compute a single “expected payoff to player 1” and optimise that; it can give a profile that neither type would actually choose.

S29. Firm 2 of type c best-responds to firm 1’s (known) quantity:

$$q_2(c) = \frac{100 - c - q_1}{2} \Rightarrow q_{2L} = \frac{90 - q_1}{2}, \quad q_{2H} = \frac{70 - q_1}{2}.$$

Firm 1 does not know which, so it responds to the *expected* rival quantity:

$$\mathbb{E}[q_2] = \frac{1}{2} \left(\frac{90 - q_1}{2} + \frac{70 - q_1}{2} \right) = 40 - \frac{q_1}{2}, \quad q_1 = \frac{100 - 20 - \mathbb{E}[q_2]}{2} = \frac{40 + q_1/2}{2}.$$

So $2q_1 = 40 + q_1/2$, giving $\frac{3}{2}q_1 = 40$ and

$$q_1 = 80/3 \approx 26.67, \quad q_{2L} = 31.67, \quad q_{2H} = 21.67.$$

Check: $\mathbb{E}[q_2] = 26.67$, and firm 1's best response to 26.67 is $(80 - 26.67)/2 = 26.67$ **ok**. Note the low-cost type produces more than firm 1 and the high-cost type less, and firm 1's quantity sits between the two complete-information answers — always state a check like this.

S30. (a) Separation requires the low type not to imitate and the high type still to participate:

$$\underbrace{1500 \geq 4000 - 250y^*}_{\text{low type stays put}} \quad \text{and} \quad \underbrace{4000 - 100y^* \geq 1500}_{\text{high type participates}}$$

which give $y^* \geq \frac{2500}{250} = 10$ and $y^* \leq \frac{2500}{100} = 25$. So $10 \leq y^* \leq 25$.

(b) The minimum is $y^* = 10$, and the **low type's no-imitation constraint** binds.

(c) With a cost of 500 the constraint becomes $1500 \geq 4000 - 500y^*$, so $y^* \geq 5$. Separation is achieved with half the education. The lesson: in a separating equilibrium the education **does nothing for productivity** — it exists only to sort — so the more expensive imitation is for the low type, the less real resource is burned. That is also the argument for cheap verification (audited records, certified tests) over expensive signals.

S31. (a) The bid is accepted iff $v \leq B$, with probability $\frac{B-5}{20}$ for $B \in [5, 25]$, and then $\mathbb{E}[v \mid v \leq B] = \frac{5+B}{2}$. Expected profit is

$$\frac{B-5}{20} \left(1.4 \cdot \frac{5+B}{2} - B \right) = \frac{B-5}{20} (3.5 - 0.3B).$$

Maximising $(B-5)(3.5 - 0.3B)$: the derivative is $3.5 - 0.3B - 0.3(B-5) = 5 - 0.6B = 0$, so

$$B^* = 8.33 \quad (\text{the formula } B^* = \frac{a}{2-k} = \frac{5}{0.6}),$$

with expected profit $\frac{3.33}{20} (3.5 - 2.5) = 0.17$.

(b) With $a = 0$ the formula gives $B^* = 0$ and the market shuts down. The reason is that the conditional mean $\mathbb{E}[v \mid v \leq B] = B/2$ means the firm is worth $1.4(B/2) = 0.7B < B$ to the buyer at *every* bid, so no positive bid is profitable. **A positive lower bound on quality is what keeps the market alive**: it guarantees that even the worst car on offer is worth something.

S32. Signalling is the *informed* party moving first with a costly, observable action — education, a warranty, a costly advertising campaign. It works only if the action is **cheaper for the high type** (the single-crossing condition); otherwise the low type imitates and the signal is uninformative.

Screening is the *uninformed* party moving first with a *menu* — insurance deductibles, airline fare classes, software tiers, quantity discounts — designed so that each type prefers its own option. It works through **incentive compatibility** plus participation, and the standard result is “no distortion at the top”: the high type gets the efficient quantity and earns an information rent, while the low type's quantity is distorted downwards.

Without a systematic difference between types in the cost of the signal or the attractiveness of the menu item, **neither mechanism can separate** and the market pools.

S33. No — the certified market unravels exactly like the uncertified one, and nobody buys the certificate.

Suppose certified cars sold at a price P . A seller certifies and sells only if the net proceeds beat keeping the car: $P - 10 \geq v$, i.e. $v \leq P - 10$. So the sellers who certify are exactly those with the *lowest* values, and their average value is $\frac{P-10}{2}$. To a buyer such a car is worth

$$1.5 \cdot \frac{P-10}{2} = 0.75(P-10) < P \quad \text{for every } P > 0,$$

so no buyer will pay P . The price collapses, and with it the incentive to certify. The key point is that the fee is **the same for every seller**, so paying it conveys no information about quality — there is no single crossing. **Only certification that actually verifies the car's condition** (or a fee that is cheaper for genuinely good sellers, which a warranty achieves) can support a separating equilibrium.

4 Part D — Auctions, securities and contests

Questions

Q34. Four risk-neutral bidders have independent private values drawn from $U[0, 1]$. (a) What is the equilibrium bid function in a first-price sealed-bid auction, and what is the intuition for the shading factor? (b) What is the seller's expected revenue in the first-price and in the second-price auction? [§7.2]

Q35. State the revenue equivalence theorem, listing its assumptions, and give three distinct circumstances in which it fails, saying in each case which format then raises more. [§7.2]

Q36. With values $U[0, 1]$ and a seller who values the object at zero, the revenue-maximising reserve price is $\frac{1}{2}$. (a) Why does it not depend on the number of bidders? (b) Why is it inefficient, and why does the seller set it anyway? [§7.3]

Q37. An oil tract is auctioned to ten firms, each with an unbiased private estimate of its common value. A manager proposes bidding the firm's own estimate. Explain what goes wrong, what the correct bid conditions on, and how the answer changes if there were only two bidders. [§7.3]

Q38. Two units are sold; bidder 1 values them at 12 and 9, bidder 2 at 10 and 7. In a uniform-price auction each bidder submits two bids, the two highest bids win, and every winning unit is paid for at the highest *rejected* bid. (a) What happens if both bid truthfully? (b) Show that bidder 1 gains by bidding (12, 0). (c) Name the phenomenon. [§7.4]

Q39. For the same two-unit setting, state whether truthful bidding is weakly dominant in the discriminatory, uniform-price and Vickrey auctions, with a one-sentence reason for each. What changes if each bidder wants only one unit? [§7.4]

Q40. A project is auctioned to *three* risk-neutral bidders whose signals v are independent $U[0, 1]$. A bidder with signal v believes the project's revenue Z is uniform on $[0, 2v]$. Bids are **debt securities**: a bid of d pledges $S(d, z) = \min\{z, d\}$ to the seller, and a winning bidder with signal v who pays according to a debt bid d earns $v - d + d^2/4v$. The highest bid wins and pays the second-highest bid. (a) Find the weakly dominant bid. (b) Does the answer depend on the number of bidders? (c) Compute the seller's expected revenue and compare it with a cash second-price auction. [§8]

Q41. In the same debt auction, explain why revenue equivalence between the first-price and second-price formats fails, and state which format the seller should prefer for debt bids. [§8.4]

Q42. Still in the debt auction, with two bidders: (a) explain why there is no dominant strategy under first-price rules; (b) write down the maximisation problem that defines the symmetric equilibrium. [§8.4]

Q43. Five risk-neutral players compete in a lottery for a prize worth 100: player i spends x_i on non-refundable tickets and wins with probability $x_i / \sum_j x_j$. (a) Find the symmetric equilibrium spend, the organiser's revenue and each player's expected profit. (b) What happens as the number of players grows? [§9]

Q44. Two firms lobby for a licence worth 100 to firm 1 and 50 to firm 2, using the same lottery technology. Find each firm's spending, the probability that firm 1 wins, and comment on which firm dissipates the larger share of its own prize. [§9]

Q45. A state must allocate one liquor licence among four bidders who each value it at 120. Compare a lottery (Tullock) with a first-price sealed-bid auction on revenue and on efficiency, and give one reason a state might nonetheless choose the lottery. [§9]

Solutions

S34. (a) Look for a symmetric increasing equilibrium $b(\cdot)$ and let a bidder of value v pretend to be v' . Since b is increasing, she wins exactly when all $n - 1$ rivals are below v' , with probability $(v')^{n-1}$, so she solves $\max_{v'} (v')^{n-1}(v - b(v'))$. Differentiating and imposing $v' = v$:

$$(n - 1)v^{n-2}(v - b) - v^{n-1}b' = 0 \Rightarrow \frac{d}{dv} [v^{n-1}b(v)] = (n - 1)v^{n-1},$$

and integrating with $b(0) = 0$ gives $v^{n-1}b = \frac{n-1}{n}v^n$, i.e. $b(v) = \frac{n-1}{n}v = 0.75v$ at $n = 4$. (Full version: §9, D1.)

Intuitively $\frac{n-1}{n}v$ is exactly $\mathbb{E}[\text{highest rival value} \mid \text{you are the highest}]$, so you bid what you expect the competition to be. Shading **falls as n rises**, because the risk of losing grows.

(b) *Second price.* Bidding v is weakly dominant, so revenue is the expected second-highest value. That density is $n(n - 1)x^{n-2}(1 - x)$ (one draw at x , one above, $n - 2$ below), giving

$$\mathbb{E}[\text{2nd}] = \int_0^1 x n(n - 1)x^{n-2}(1 - x)dx = \frac{n - 1}{n + 1} = \frac{3}{5} = 0.6.$$

First price. The winner is the highest bidder and pays $\frac{n-1}{n}$ of her value, and $\mathbb{E}[\max] = \int_0^1 x nx^{n-1}dx = \frac{n}{n+1}$, so revenue is $\frac{n-1}{n} \cdot \frac{n}{n+1} = \frac{n-1}{n+1} = 0.6$ again. The two shadings cancel exactly — that is revenue equivalence, and it is worth showing both integrals rather than asserting the theorem. (§9, D2–D3.)

S35. Revenue equivalence: with (i) risk-neutral bidders, (ii) independent private values drawn from the same distribution, (iii) an allocation rule that always gives the object to the highest bidder, and (iv) a bidder with the lowest possible value earning zero expected surplus, every standard auction format yields the *same* expected revenue.

Three failures:

- **Risk-averse bidders** — they shade less in a first-price auction to reduce the risk of losing, so the **first-price** auction raises more.
- **Common or affiliated values** — in an ascending auction bidders learn from each other's drop-out points, which reduces the winner's curse and lets them bid more (the linkage principle), so the **English/second-price** format raises more.

- **Bids in securities rather than cash** — with debt bids, second-price raises more than first-price (Q40–Q41).

Other valid answers: asymmetric bidders, budget constraints, and collusion, which is easier to sustain in ascending auctions.

S36. (a) *The computation.* Expected revenue is the reserve paid when exactly one bidder clears it, plus the second-highest value when two or more do:

$$\Pi(r) = r \cdot nr^{n-1}(1-r) + \int_r^1 y n(n-1)y^{n-2}(1-y) dy,$$

which differentiates to $\Pi'(r) = nr^{n-1}(1-2r)$, zero at $r^* = 1/2$ for every n . *The argument to write if you are short of time.* The reserve trades off two things: the probability that no sale occurs against the higher price when a sale does occur. Both effects concern only the *marginal* buyer whose value sits at the reserve, and the reserve binds only when *every* bidder's value is below it — the comparison at that margin is between the seller's own value (0) and the reserve, which does not involve n . So $r^* = 1/2$ whatever the number of bidders.

(b) It is **inefficient** because with probability r^n nobody wins even though every bidder values the object above the seller. The seller sets it anyway because it **raises the price in the much more likely event that a sale does occur**, and she cares about revenue rather than surplus. As n grows the reserve almost never binds, so the loss vanishes while the gain does too.

S37. Bidding your own estimate is the **winner's curse**. Winning means your estimate was the *highest of ten*, which is systematically optimistic: $\mathbb{E}[v \mid s_i, s_i = \max] < \mathbb{E}[v \mid s_i]$. So a bidder who wins at her own estimate has, on average, overpaid.

The correct bid conditions on the event of winning: bid $\mathbb{E}[v \mid s_i \text{ is the highest of } n]$, and then shade further for the first-price rule. The shading must **increase with n** : “I had the highest of ten signals” is far worse news than “I had the higher of two.” With only two bidders the curse is much weaker, so the correct bid is considerably **closer to the raw estimate**. Say this explicitly — it is the part students omit.

S38. (a) Truthful bids are 12, 9 from bidder 1 and 10, 7 from bidder 2. The two highest bids, 12 and 10, win one unit each, and the highest rejected bid is 9, so both winners pay 9. Bidder 1's surplus is $12 - 9 = 3$.

(b) If bidder 1 bids (12, 0) the bid list is 12, 0, 10, 7: the winners are still 12 and 10, but the highest rejected bid is now 7, so the price falls to 7 and bidder 1's surplus rises to $12 - 7 = 5$. She gave up a unit she was never going to win anyway and bought herself a lower price on the unit she does win.

(c) **Demand reduction.** It is why uniform-price auctions are not strategy-proof and why bidding one's true values is not weakly dominant there.

S39. Discriminatory (pay-as-bid): no. You pay your own bid, so you always shade, exactly as in a single-unit first-price auction. **Uniform price: no.** Your later bids can set the price you pay on your earlier units, so demand reduction pays (Q38). **Vickrey: yes.** A bidder winning k units pays the sum of the k highest rejected bids *submitted by others*, so no bid of yours ever affects what you pay — only whether you win. That restores the second-price logic unit by unit. With **unit demand**, the uniform-price auction *is* a second-price auction, so bidding your value becomes weakly dominant there too.

S40. (a) Apply the second-price principle: **bid the amount that makes your payoff on winning exactly zero**, because that is what makes both over- and under-bidding harmless. Setting $v - d + \frac{d^2}{4v} = 0$ gives $4v^2 - 4vd + d^2 = 0$, i.e. $(2v - d)^2 = 0$, so

$$d(v) = 2v.$$

(b) **No.** Weak dominance is a statement about *every* profile of rival bids, so it holds for any number of bidders; the case-by-case proof (comparing d' with d^* against the highest rival bid $\hat{d}$) never uses n . What does change with n is the *revenue*, because the second-highest signal rises.

(c) With three bidders, condition on the winner's signal v . The second-highest signal w is the maximum of two independent draws from $U[0, v]$, so its density is $2w/v^2$ and $\mathbb{E}[w] = \frac{2v}{3}$, $\mathbb{E}[w^2] = \frac{v^2}{2}$. The winner pays

$$\mathbb{E}[S(z, 2w)] = 2w - \frac{(2w)^2}{4v} = 2w - \frac{w^2}{v} \Rightarrow \mathbb{E} = 2 \cdot \frac{2v}{3} - \frac{v^2/2}{v} = \frac{4v}{3} - \frac{v}{2} = \frac{5v}{6}.$$

The highest of three uniforms has density $3v^2$, so expected revenue is $\int_0^1 \frac{5v}{6} 3v^2 dv = 5/8 = 0.625$.

In a *cash* second-price auction a bidder with signal v values the project at $\mathbb{E}[Z \mid v] = v$, so she bids v and the winner pays the second-highest signal: $\mathbb{E}[w \mid v] = \frac{2v}{3}$, and unconditionally $\int_0^1 \frac{2v}{3} 3v^2 dv = 1/2$. **The debt auction raises 25% more.** The economics: a debt security pays the seller more in exactly the states where the project succeeds, so it extracts part of the bidder's informational rent that cash cannot reach.

S41. Revenue equivalence rests on the envelope argument that a bidder's equilibrium surplus satisfies $U'(v) = \Pr(\text{win at } v)$, which requires the payoff to be *linear* in the type. Here the type v enters twice — once as the value and once through the *distribution of Z* , since $Z \sim U[0, 2v]$ — so the payoff $v - d + d^2/4v$ is not linear in v and the envelope

condition becomes $U'(v) = \Pr(\text{win})[1 - d^2/4v^2]$. Once that term differs across formats, **the two auctions no longer raise the same revenue.**

Concretely, with two bidders the second-price debt auction raises $4/9 \approx 0.444$, the first-price debt auction $10/27 \approx 0.370$, and either cash auction $1/3$. So for debt — a “sub-convex” security — **the seller should prefer the second-price format**, and both debt formats beat cash.

S42. (a) Under first-price rules **what you pay is your own bid**, so the trade-off between winning more often and paying more depends entirely on how aggressively the rivals bid. There is no single bid that is optimal against every rival bid, so no dominant strategy exists — exactly as in a first-price cash auction. (Under second-price rules your bid never affects your price, which is what makes dominance possible there.)

(b) Assume a symmetric, strictly increasing equilibrium $d(\cdot)$. A bidder of type v who bids as if her type were v' wins whenever v' exceeds the rival's signal, which with one rival and uniform signals happens with probability v' . So she solves

$$\max_{v'} v' \left(v - d(v') + \frac{d(v')^2}{4v} \right),$$

and the equilibrium is the fixed point where $v' = v$ is optimal. (Solving it: the first-order condition reduces to $d'(v) = 1 - \frac{d(v)}{2v}$, whose solution with $d(0) = 0$ is $d(v) = \frac{2}{3}v$.)

S43. (a) Player i maximises $\frac{x_i}{x_i + X_{-i}}V - x_i$, with first-order condition $\frac{X_{-i}}{(x_i + X_{-i})^2}V = 1$. Imposing symmetry ($X_{-i} = (n-1)x$):

$$\frac{(n-1)x}{(nx)^2}V = 1 \Rightarrow x^* = \frac{(n-1)V}{n^2} = \frac{4(100)}{25} = \underline{16}, \quad \text{revenue} = \frac{(n-1)V}{n} = \underline{80}, \quad \text{profit} = \frac{V}{n^2} = \underline{4} \text{ each.}$$

Check: each player wins with probability $1/5$ for an expected prize of 20 and spends 16, leaving 4 **ok**.

(b) Total spending is the fraction $(n-1)/n$ of the prize and **tends to the whole prize** as n grows: complete **rent dissipation**. Individual profit V/n^2 collapses even faster. This is the standard argument that competitive rent-seeking destroys most of the value it competes for.

S44. Derive it rather than quoting it. The two first-order conditions are $\frac{x_2}{(x_1 + x_2)^2}V_1 = 1$ and $\frac{x_1}{(x_1 + x_2)^2}V_2 = 1$. They share the denominator, so dividing one by the other kills it:

$$\frac{x_2V_1}{x_1V_2} = 1 \Rightarrow \frac{x_1}{x_2} = \frac{V_1}{V_2},$$

so efforts are proportional to prizes and player i wins with probability $V_i/(V_1 + V_2)$. Substituting $x_2 = x_1V_2/V_1$ back into the first condition gives $x_i = \frac{V_i^2V_j}{(V_i + V_j)^2}$ (S9, D8). With $V_1 = 100$, $V_2 = 50$:

$$x_1 = \frac{100^2 \cdot 50}{150^2} = \underline{22.22}, \quad x_2 = \frac{50^2 \cdot 100}{150^2} = \underline{11.11}.$$

Firm 1 wins with probability $\frac{22.22}{33.33} = \underline{2/3}$.

The stronger firm spends more in absolute terms, but as a *share of its own prize* firm 1 spends $22.22/100 = 22\%$ and firm 2 spends $11.11/50 = 22\%$ — equal here. The sharper general point to make: the **underdog is deterred rather than encouraged** by a larger gap, and if V_2 were small enough relative to V_1 it would optimally drop out altogether, so always check for that corner.

S45. Lottery: with $n = 4$ and $V = 120$, each spends $\frac{3(120)}{16} = 22.5$ and the state collects 90; each bidder keeps $120/16 = 7.5$. *First-price auction with complete information and equal values*: competition drives the price to the common value, so the state collects 120 and the bidders earn nothing.

On **revenue** the auction wins (120 against 90). On **efficiency** the auction also wins once values differ, because it always gives the licence to the highest-value bidder, whereas the lottery gives it to a lower-value bidder with positive probability — and in addition the lottery burns 90 of real resources in ticket-buying, which is pure waste.

A state might still prefer the lottery because it **gives smaller or credit-constrained bidders a genuine chance**, which may be a distributional or political objective; because an auction among very asymmetric bidders is a foregone conclusion and deters participation; or because auction revenue is politically harder to justify than a lottery.

5 Part E — Matching, market design and cooperative games

Questions

Q46. Three applicants and three firms have the preferences below (best first).

a_1 :	$f_1 \succ f_2 \succ f_3$	f_1 :	$a_2 \succ a_3 \succ a_1$
a_2 :	$f_2 \succ f_3 \succ f_1$	f_2 :	$a_3 \succ a_1 \succ a_2$
a_3 :	$f_3 \succ f_1 \succ f_2$	f_3 :	$a_1 \succ a_2 \succ a_3$

(a) Run applicant-proposing deferred acceptance. (b) Run firm-proposing deferred acceptance. (c) Find every stable matching. (d) What does the comparison illustrate? [§10.2]

Q47. Prove that the matching produced by deferred acceptance is stable. [§10.2]

Q48. State, with a one-line reason each: (a) whether applicant-proposing DA is strategy-proof for applicants; (b) whether it is strategy-proof for firms; (c) whether *any* stable mechanism can be strategy-proof for everyone. [§10.2]

Q49. Four agents own one house each and have the preferences: agent 1 prefers $h_2 \succ h_3 \succ h_1 \succ h_4$; agent 2 prefers $h_1 \succ h_4 \succ h_2 \succ h_3$; agent 3 prefers $h_4 \succ h_1 \succ h_3 \succ h_2$; agent 4 prefers $h_3 \succ h_2 \succ h_4 \succ h_1$. Run Top Trading Cycles and state two properties of the outcome. [§10.3]

Q50. Compare student-proposing DA, TTC and the Boston mechanism on stability, efficiency for students, and strategy-proofness. Which trade-off is unavoidable, and why does the Boston mechanism penalise honest families? [§10.3]

Q51. Show by example that a stable matching need not exist in the roommates problem, and explain what feature of the two-sided problem Gale–Shapley relies on. [§10.2]

Q52. A three-player game has $v(\{i\}) = 0$ for each i , $v(\{1, 2\}) = 60$, $v(\{1, 3\}) = 80$, $v(\{2, 3\}) = 100$ and $v(N) = 120$. (a) Find the core. (b) Find the Shapley value. (c) Is the Shapley value in the core? Comment. [§11.1, §11.4]

Q53. A landowner employs $n = 4$ workers, with $f(1) = 20$, $f(2) = 34$, $f(3) = 44$, $f(4) = 50$ and $f(5) = 54$. (a) Verify that marginal product is decreasing. (b) Find the core. (c) What is the most the workers can obtain in total? (d) The workers unionise and forbid any group of fewer than four from working with the landowner. What is the core now? [§11.2]

Q54. State, with one sentence of justification each: (a) whether the core can be empty; (b) whether the Shapley value can be outside the core; (c) which class of games guarantees that it is inside. Then compute the Shapley–Shubik power index for shareholders holding 50, 40 and 10 votes with a quota of 51, and comment. [§11.3–11.4]

Solutions

S46. (a) *Applicant-proposing.* Round 1: $a_1 \rightarrow f_1$, $a_2 \rightarrow f_2$, $a_3 \rightarrow f_3$. Each firm receives exactly one proposal, so each holds it and there are no rejections. The algorithm stops:

$$M_A = \{a_1f_1, a_2f_2, a_3f_3\} \quad \text{— every applicant gets her first choice.}$$

(b) *Firm-proposing.* Round 1: $f_1 \rightarrow a_2$, $f_2 \rightarrow a_3$, $f_3 \rightarrow a_1$. Again one proposal each, all held:

$$M_F = \{a_2f_1, a_3f_2, a_1f_3\} \quad \text{— every firm gets its first choice.}$$

(c) Checking all six matchings for blocking pairs gives **three stable matchings**: M_A , M_F , and the “middle” matching $\{a_1f_2, a_2f_3, a_3f_1\}$ in which everyone — applicant and firm alike — receives their *second* choice.

(d) The set of stable matchings has a **lattice** structure with the two sides’ interests exactly opposed: the applicant-proposing outcome is the *best* stable matching for every applicant and simultaneously the *worst* for every firm, and vice versa. **Who proposes matters**, and the middle matching shows that the two extremes are not the only stable options.

S47. Let M be the DA outcome and suppose, for contradiction, that (a, f) is a blocking pair: applicant a prefers f to her assignment, and firm f prefers a to its assignment.

Since a prefers f to what she ended up with, and applicants propose in order of preference, a **must have proposed to f at some earlier round**. She is not matched to f , so f **rejected** her. A firm rejects a proposal only when it is already holding an applicant it prefers. Moreover a firm’s holding **only improves over time**, because in each round it keeps the best of its current holding and the new proposals.

Therefore the applicant f ends with is at least as good, in f ’s ranking, as the one it was holding when it rejected a — which it already preferred to a . So f prefers its final partner to a , contradicting the assumption that (a, f) blocks. Individual rationality holds because nobody ever proposes to, or holds, an unacceptable partner. Hence M is stable. $\square$

S48. (a) **Yes** (Dubins–Freedman, Roth): DA already gives each proposer the best partner she has in *any* stable matching, so there is nothing left to gain; a misreport can only get her rejected earlier by someone she could have had.

(b) **No**: a firm can profitably *truncate* its list — declaring an acceptable applicant unacceptable — so that the applicant it really wants is not taken by a rival.

(c) **No** (Roth 1982 impossibility): no stable matching mechanism is strategy-proof for both sides, and a two-by-two example with opposed preferences already produces a profitable manipulation. This is the single most quotable result in the topic.

S49. *Round 1.* Agent 1 points at h_2 (owner 2); agent 2 points at h_1 (owner 1). That is a cycle $1 \rightarrow 2 \rightarrow 1$, so assign h_2 to agent 1 and h_1 to agent 2, and remove them. *Round 2.* Only h_3 and h_4 remain. Agent 3’s favourite among them is h_4

(owner 4); agent 4's favourite is h_3 (owner 3). Cycle again: assign h_4 to agent 3 and h_3 to agent 4.

Outcome: $1 \rightarrow h_2, 2 \rightarrow h_1, 3 \rightarrow h_4, 4 \rightarrow h_3$.

Two properties: the outcome is **Pareto efficient** and **strategy-proof** (truthful reporting is dominant); in a housing market it is also the **unique core allocation**, so no group of agents could re-trade among themselves and all do better.

S50.

	Stable / no justified envy	Pareto efficient for students	Strategy-proof for students
Student-proposing DA	yes	no	yes
TTC	no	yes	yes
Boston	no	(efficient only given strategic play)	no

The unavoidable trade-off is **between stability and student-side Pareto efficiency**: stability protects priorities, and respecting a priority sometimes blocks a trade that would make two students better off and nobody worse off. TTC takes the efficiency side and accepts justified envy; DA takes the stability side and accepts inefficiency.

Boston penalises honesty because **first-choice priority is spent at the first round and lost if you miss**. A family that truthfully ranks a popular school first and fails to get in finds that the second-choice school's seats have already gone to families who listed it first. The safe strategy is to rank a school you are sure of, so sophisticated families do better than equally-placed naive ones — an equity objection as much as an efficiency one.

S51. Take three people a, b, c (and a fourth, d , whom everyone ranks last) with

$$a : b \succ c \succ d, \quad b : c \succ a \succ d, \quad c : a \succ b \succ d.$$

Any matching pairs one of a, b, c with d and the other two together. Say b and c are paired and a is with d . Then c prefers a to b , and a prefers anyone to d , so (a, c) blocks. By symmetry the same happens in each of the three cases, so **no stable matching exists**.

Gale–Shapley relies on **two-sidedness**: proposals always flow from one fixed side to the other, so a firm's holding improves monotonically and the process cannot cycle. In the roommates problem everyone is both a proposer and a receiver, the monotonicity argument breaks, and the preference cycle above has nowhere to resolve.

S52. (a) Efficiency gives $x_1 + x_2 + x_3 = 120$. Rewrite each pairwise constraint as a ceiling on the player left out:

$$x_1 \leq 120 - 100 = 20, \quad x_2 \leq 120 - 80 = 40, \quad x_3 \leq 120 - 60 = 60,$$

together with $x_i \geq 0$. The ceilings sum to $20 + 40 + 60 = 120 = v(N)$, so every player must be *at* her ceiling:

the core is the single point $(20, 40, 60)$.

(b) Averaging marginal contributions over the six orders gives

$$\varphi_1 = \underline{30}, \quad \varphi_2 = \underline{40}, \quad \varphi_3 = \underline{50} \quad (\text{sum } 120 \text{ ok}).$$

(c) **No**. The Shapley value gives player 1 thirty, but the core caps her at twenty, so the coalition $\{2, 3\}$ — worth 100 — would receive only $40 + 50 = 90$ and would walk out. The comparison is the cleanest statement of the difference between the two concepts: the **Shapley value asks what is fair** (player 1 does contribute 30 on average) while the **core asks what is stable** (player 1 is the least essential and the others can do without her).

S53. (a) Marginal products are 20, 14, 10, 6, 4 — **strictly decreasing ok**.

(b) Each worker's ceiling is $f(5) - f(4) = 54 - 50 = 4$, with a floor of 0, and the landowner takes the residual:

$$0 \leq x_i \leq 4 \text{ for each of the four workers,} \quad x_1 = 54 - \sum_{i \geq 2} x_i \in [38, 54].$$

(c) At most $4 \times 4 = 16$, leaving the landowner 38.

(d) Now the only productive coalitions are the grand coalition (54) and the landowner *alone* ($f(1) = 20$) — every intermediate group is forbidden to produce. The core becomes

$$\{x : x_i \geq 0 \text{ for workers, } x_1 \geq 20, \sum x_i = 54\},$$

so the workers can now collectively obtain up to 34 rather than 16. **Unionisation works by destroying the landowner's outside option** — the coalition that used to cap wages at the last worker's marginal product is no longer legal. Note it is *not* true that every allocation is now in the core: the landowner can still work alone.

S54. (a) **Yes**. The three-person majority game, where any two players can split 1, has an empty core: adding the three pairwise constraints gives $2 \sum x_i \geq 3$ while efficiency requires $\sum x_i = 1$.

(b) **Yes** — Q52 is an example, as is the glove market, where the core gives the scarce side everything while the Shapley value pays the long side $1/6$ each.

(c) **Convex games**, in which a player's marginal contribution grows with the size of the coalition she joins; there the Shapley value always lies in the core.

Power index. Write out the six arrival orders and mark who first pushes the running total past 51. A (50 votes) is pivotal in four orders (BAC , BCA , CAB , CBA); B is pivotal in ABC ; C is pivotal in ACB . So

$$\varphi_A = 2/3, \quad \varphi_B = 1/6, \quad \varphi_C = 1/6.$$

Power is not proportional to votes: the 40% shareholder has exactly the same power as the 10% shareholder, because whenever A is absent from the front of the queue either of them completes a winning coalition equally often. A shareholder buying votes should care about crossing pivotal thresholds, not about the raw percentage.

6 Part F — Risk, insurance, search and other solution concepts

Questions

Q55. A decision maker with $u(x) = x^{1/3}$ faces a 50–50 gamble between 8 and 216. Find the expected value, the certainty equivalent and the risk premium, and say what the sign of the premium tells you. [§12.2]

Q56. Wealth is 100; with probability 0.5 a loss of 64 occurs; utility is $\sqrt{W}$. Find (a) the actuarially fair premium for full cover, (b) the maximum premium the person would pay, (c) the risk premium, and (d) a monopoly insurer's profit per customer. Verify that (b) = (a) + (c). [§12.3]

Q57. Two customers each have wealth 10,000 and each faces a 50% chance of losing 9,600. One has utility $W^{1/2}$ and the other $W^{1/4}$, but you cannot tell which is which. You are risk neutral and may make a single take-it-or-leave-it full-cover offer. (a) Find each customer's maximum premium and say which is more risk averse. (b) What single premium maximises your profit? (c) You can pay 400 to learn who is who and then price separately. Should you? [§12.5]

Q58. A project pays +150 with probability 0.5 and −90 otherwise; the alternative pays 0 for certain, and the firm is risk neutral. (a) What does it do without information, and what is the expected payoff? (b) What is the expected value of perfect information? (c) Why must imperfect information be worth strictly less? [§12.6]

Q59. A person with $u(W) = W^2$ has wealth 4 and is offered a fair coin paying +2 or −2, with a participation fee F payable whatever the outcome. Find the largest fee she will accept, and comment on the size of the answer. [§12.4]

Q60. Using the numbers of Q56, decide whether the person buys full cover (a) at the actuarially fair premium and (b) at a premium loaded 20% above fair. State the general result. [§12.3]

Q61. A known option is worth 5. Box A pays 40 with probability 0.25; box B pays 15 with probability 0.8. Opening either box costs 2. (a) Compute the reservation values. (b) State the optimal policy completely. (c) Compute its expected payoff and compare with opening B first. [§13]

Q62. Job offers are drawn independently from $U[0, 80]$ and each draw costs 1.6. Find the reservation wage, and state what happens to it if (i) the search cost rises, (ii) the offer distribution becomes more dispersed. [§13]

Q63. Describe the optimal policy in the secretary problem and its success probability, and say what feature of a question tells you to use it. [§13]

Q64. In Chicken with payoffs (0, 0) for (D, D) , (7, 2) for (D, C) , (2, 7) for (C, D) and (6, 6) for (C, C) , a mediator privately recommends (C, C) , (D, C) and (C, D) with probability $\frac{1}{3}$ each. (a) Verify that obedience is optimal. (b) Compute the expected payoff and compare with the mixed Nash equilibrium. (c) What breaks if the mediator's draw is announced publicly? [§14]

Q65. In Hawk–Dove the prize is V and fighting costs C . (a) Find the evolutionarily stable strategy when $V = 6$ and $C = 15$. (b) What is the ESS when $V = 10$ and $C = 4$? (c) State the algebraic warning. [§14]

Q66. Players guess a number in $[0, 100]$ and the winner is closest to $\frac{3}{4}$ of the average. Assuming level-0 players randomise uniformly, compute the level-1, level-2 and level-3 choices and the Nash equilibrium. Why must you state the level-0 assumption? [§14]

Q67. Three thousand drivers travel from S to E . Route 1 is $S \rightarrow X$ taking $T/100$ minutes followed by $X \rightarrow E$ taking 40; route 2 is $S \rightarrow Y$ taking 40 followed by $Y \rightarrow E$ taking $T/100$. (a) Find the equilibrium travel time. (b) A free, instantaneous road $X \rightarrow Y$ opens. Find the new equilibrium time and verify it is an equilibrium. (c) Name the paradox and give the policy implication. [§14]

Solutions

S55. $EX = \frac{1}{2}(8) + \frac{1}{2}(216) = 112$. Expected utility is $\frac{1}{2}(8^{1/3}) + \frac{1}{2}(216^{1/3}) = \frac{1}{2}(2) + \frac{1}{2}(6) = 4$, so

$$CE = 4^3 = 64, \quad \text{risk premium} = 112 - 64 = 48.$$

A **positive** risk premium means the person is risk averse: she would accept a certain 64 rather than a gamble worth 112 on average, and would pay up to 48 to be rid of the risk. The premium here is very large relative to the mean because $x^{1/3}$ is strongly concave and the gamble is very spread out.

S56. The two states are $W = 100$ and $W = 36$.

(a) Fair premium $= \pi L = 0.5(64) = 32$.

(b) $\mathbb{E}u = 0.5(10) + 0.5(6) = 8$, so CE = 64 and the maximum premium is $100 - 64 = 36$.

(c) $\mathbb{E}W = 0.5(100) + 0.5(36) = 68$, so the risk premium is $68 - 64 = 4$.

(d) A monopolist charges 36 and expects to pay out 32, so profit is 4 per customer — exactly the risk premium, which is the entire gain from trade.

Check: $36 = 32 + 4$ ok.

S57. The two states are 10,000 and 400.

(a) $\sqrt{\cdot}$ type: $\mathbb{E}u = 0.5(100) + 0.5(20) = 60$, so CE = 3,600 and $P^{\max} = 6,400$. $W^{1/4}$ type: $\mathbb{E}u = 0.5(10) + 0.5(4.4721) = 7.2361$, so CE = $7.2361^4 = 2,741$ and $P^{\max} = 7,259$. The $W^{1/4}$ **customer is more risk averse** — a smaller exponent in the CRRA family means more concavity — and correspondingly pays more.

(b) Your expected payout per customer is $0.5(9,600) = 4,800$. Charging 6,400: *both* accept, profit = $2(6,400 - 4,800) = 3,200$. Charging 7,259: only the more risk-averse one accepts, profit = $7,259 - 4,800 = 2,459$. So offer **6,400**. A single price must be acceptable to the *least* willing customer you want to keep — the same logic as the entry fee in a two-part tariff.

(c) With tailored deals, profit = $6,400 + 7,259 - 2(4,800) - 400 = 3,459 > 3,200$, so **yes, pay the 400**. The information is worth 659 gross and 259 net. Note the general principle: information has value only because it *changes what you do* — here it lets you stop leaving 859 of surplus with the more risk-averse customer.

S58. (a) $\mathbb{E} = 0.5(150) + 0.5(-90) = 75 - 45 = 30 > 0$, so the firm **invests**, with expected payoff 30.

(b) With perfect information it invests only in the good state: $\mathbb{E} = 0.5(150) + 0.5(0) = 75$. So

$$EVPI = 75 - 30 = 45,$$

which is exactly the avoided loss $0.5(90)$ — a useful check.

(c) Imperfect information sometimes points the wrong way, so the firm occasionally still invests in the bad state or skips the good one. Formally the decision rule based on a noisy signal is one of the feasible rules available to a decision maker with perfect information, so the perfect-information value is an **upper bound**. An extreme case makes it vivid: a completely uninformative signal is worth zero.

S59. Without the gamble, utility is $4^2 = 16$. With fee F the two states are $6 - F$ and $2 - F$, so participation requires

$$0.5(6 - F)^2 + 0.5(2 - F)^2 \geq 16.$$

Expanding: $0.5[36 - 12F + F^2 + 4 - 4F + F^2] = 20 - 8F + F^2 \geq 16$, i.e. $F^2 - 8F + 4 \geq 0$, whose roots are $F = 4 \pm \sqrt{12} = 4 \pm 3.464$. The relevant (small) root gives

$$F \leq 4 - 3.464 = 0.536.$$

Comment: she is risk *loving*, so she will indeed pay for a fair gamble — but only about 0.54 on wealth of 4, i.e. 13% of wealth for a gamble that swings her wealth by $\pm 50\%$. **Even a risk lover pays only a modest amount for a small fair gamble**, which is why casinos rely on volume and on an unfair house edge rather than on large participation fees.

S60. (a) At the fair premium of 32, insured wealth is $100 - 32 = 68$ for certain, worth $\sqrt{68} = 8.246$, against uninsured expected utility of 8. Since $8.246 > 8$, **she buys**. This is general: at an actuarially fair price a risk-averse person always buys *full* cover, because fair insurance transfers wealth between states at exactly the probability ratio, and the optimum then equalises marginal utility across states, which means equal wealth.

(b) A 20% load gives a premium of $1.2(32) = 38.4$, which exceeds her maximum of 36, so **she refuses full cover**. The general result: at a loaded price a risk-averse person buys **strictly less than full cover**, since each extra rupee of coverage now costs more than its actuarial value. That is why real policies carry deductibles and co-payments.

S61. (a) $r_A = 40 - \frac{2}{0.25} = 32$ and $r_B = 15 - \frac{2}{0.8} = 12.5$. Both exceed the 5 already in hand, so investigating is worthwhile.

(b) Since $r_A > r_B$, **open box A first**. If A pays 40, **stop and take it** — 40 exceeds the highest remaining reservation value $r_B = 12.5$. If A fails, the best in hand is 5, which is below r_B , so **open box B**; take 15 if it pays, otherwise take the known 5.

(c) Expected payoff:

$$0.25(40 - 2) + 0.75[0.8(15 - 4) + 0.2(5 - 4)] = 9.5 + 0.75(9) = 16.25.$$

Opening B first: if B pays 15, that is still below $r_A = 32$, so you would open A anyway, giving $0.25(36) + 0.75(11) = 17.25$; if B fails you open A, giving $0.25(36) + 0.75(1) = 9.75$. Total $0.8(17.25) + 0.2(9.75) = 15.75$, which is 0.5 worse.

Box A goes first despite having the lower expected value (10 against 12), because its success would end the search immediately — reservation values price that option, expected values do not.

S62. The reservation wage solves $c = \mathbb{E}[\max(w - w^*, 0)] = \frac{(80 - w^*)^2}{160}$. With $c = 1.6$:

$$(80 - w^*)^2 = 256 \Rightarrow 80 - w^* = 16 \Rightarrow w^* = 64.$$

(i) A **higher search cost lowers w^*** — each extra draw is dearer, so you settle sooner. (ii) A **more dispersed distribution raises w^*** , because the option value of another draw comes from the upper tail. The policy corollary: unemployment benefits act like a lower search cost, raising the reservation wage and lengthening unemployment spells — a standard exam application.

S63. Reject the first $n/e \approx 37\%$ of candidates outright, then accept the first candidate who is better than everyone seen so far. This selects the very best candidate with probability approximately $1/e \approx 37\%$, and remarkably that probability does not fall as n grows.

The tell-tale features are **irrevocable decisions, no recall, and only relative ranks observable**. Whenever a question says you must accept or reject on the spot and cannot go back, the answer has this shape: a **learning phase** that sets the benchmark, followed by a **threshold phase** that takes the first record-breaker.

S64. (a) Told “D”. That recommendation occurs only in the profile (D, C) , so the player infers the opponent was told C with probability 1. Obeying pays 7; deviating to C pays 6. **Obey.** Told “C”. Two profiles contain a C for her, (C, C) and (C, D) , each with prior $\frac{1}{3}$, so conditionally each has probability $\frac{1}{2}$. Obeying pays $\frac{1}{2}(6) + \frac{1}{2}(2) = 4$; deviating to D pays $\frac{1}{2}(7) + \frac{1}{2}(0) = 3.5$. **Obey.**

(b) Expected payoff $= \frac{1}{3}(6) + \frac{1}{3}(7) + \frac{1}{3}(2) = 5$ each. The mixed Nash equilibrium has each player daring with probability $\frac{1}{3}$ (from $7 - 7p = 6 - 4p$) and pays $7 - 7(\frac{1}{3}) = 14/3 \approx 4.67$. So the device makes *both* players better off, and it does so by **never recommending the crash (D, D)** , which independent mixing cannot avoid.

(c) If the draw were public, each player would know the other’s recommendation, so the only self-enforcing recommendations would be Nash equilibria of the underlying game — the device would collapse to a **public coin flip between the two pure equilibria**, worth 4.5 each. **The gain comes from the signals being private.**

S65. (a) With $C > V$ there is no pure ESS: Hawk fails because $u(H, H) = \frac{V-C}{2} < 0 = u(D, H)$, and Dove fails because $u(D, D) = \frac{V}{2} < V = u(H, D)$. The mixed ESS has a hawk share

$$p^* = \frac{V}{C} = \frac{6}{15} = 0.4.$$

(b) With $V = 10 > C = 4$, $u(H, H) = \frac{10-4}{2} = 3 > 0 = u(D, H)$, so Hawk is a strict best reply to itself and **“always Hawk” is a pure ESS**: the population is entirely aggressive.

(c) **The formula $p^* = V/C$ is valid only when $C > V$.** Applying it blindly to $V = 10$, $C = 4$ gives $p^* = 2.5$, which is not a probability — the algebraic signal that the interior solution has left the unit interval and the answer is a corner.

S66. Level-0 randomises uniformly on $[0, 100]$, so its mean is 50. Then

$$L1 = \frac{3}{4}(50) = 37.5, \quad L2 = \frac{3}{4}(37.5) = 28.125, \quad L3 = \frac{3}{4}(28.125) = 21.09,$$

and the Nash equilibrium is the only fixed point of $x = \frac{3}{4}x$, namely **0** (reached by iterated deletion: nobody should ever choose above $\frac{3}{4}$ of the current upper bound).

You must state the level-0 assumption because **it is the only input that is not pinned down by the model**, and the whole ladder shifts with it: if level-0 were assumed to choose 100 rather than to randomise, the ladder would read 75, 56.25, 42.19. Examiners award the mark for making the assumption explicit.

S67. (a) By symmetry 1500 drivers take each route, so each link carries 1500 and the travel time is $\frac{1500}{100} + 40 = 55$ minutes. It is an equilibrium because switching would make your new congestible link carry 1501 cars, taking $15.01 + 40 > 55$.

(b) With the free link, consider everyone taking $S \rightarrow X \rightarrow Y \rightarrow E$. Both congestible links then carry 3000 cars, so the time is $30 + 0 + 30 = 60$ minutes. It is an equilibrium: a deviator to $S \rightarrow X \rightarrow E$ faces $30 + 40 = 70$, and to $S \rightarrow Y \rightarrow E$ faces $40 + 30 = 70$, both worse than 60. So everyone stays, and **every driver is 5 minutes worse off than before the road was built**.

(c) **Braess’s paradox.** The new link is individually attractive but funnels everyone onto *both* congestible links, and each driver ignores the delay she imposes on others — a congestion externality. The policy implication is that **adding capacity to a congested network can slow it down, and closing roads can speed it up** (documented in Seoul, New York and Stuttgart); capacity decisions must be evaluated at the new equilibrium, not at current flows.

7 Part G — Monopoly, oligopoly and pricing

Questions

Q68. A monopolist faces $P = 150 - 3Q$ with $C(Q) = 100 + 30Q$. Find (a) price, quantity and profit; (b) the Lerner index and the elasticity at the optimum, verified two ways; (c) the deadweight loss; (d) what happens if the fixed cost rises to 900; (e) the revenue-maximising quantity, and why it differs. [§15.1–15.3]

Q69. A monopolist faces $P = 120 - Q$ and has two plants with $MC_1 = 4q_1$ and $MC_2 = 2q_2$. Find total output, the split, and the price. Verify the optimality condition. [§15.4]

- Q70.** Demand is $Q = 120 - P$ and costs are $C(Q) = 900 + 20Q$. (a) Is this a natural monopoly? (b) Find the unregulated price and profit. (c) Find the marginal-cost-pricing outcome and say whether it is viable. (d) Find the average-cost price. [§15.5]
- Q71.** Say whether each is true or false, with one sentence: (i) in long-run monopolistic competition, firms produce at minimum average cost; (ii) zero economic profit implies no deadweight loss; (iii) price exceeds both the short-run and the long-run minimum average cost. [§15.6]
- Q72.** Inverse demand is $P = 120 - Q$; firm 1's marginal cost is 20 and firm 2's is 50. (a) Find Cournot quantities, price and profits. (b) Redo it with firm 2's cost at 90 and explain the answer carefully. [§16.1]
- Q73.** With the same demand and both costs equal to 20, find the Stackelberg outcome and compare the leader's profit with her Cournot profit. Then say whether the leader's quantity is a best response to the follower's, and why that matters. [§16.2]
- Q74.** Two firms with differentiated products face $q_1 = 50 - 3p_1 + 2p_2$ and $q_2 = 50 - 3p_2 + 2p_1$, with marginal cost 5. Find (a) the simultaneous-move prices and profits; (b) the outcome when firm 1 moves first; (c) who gains from sequencing. [§16.4]
- Q75.** Market demand is $Q = 200 - 2P$; a competitive fringe supplies $Q^f = 3P - 30$ for $P \geq 10$; the dominant firm has constant $MC = 10$. Find the price, the dominant firm's output and the fringe's output. [§16.5]
- Q76.** Above a current price of 30, a firm's demand is $Q = 50 - P$; below it, rivals match and demand is $Q = 30 - \frac{1}{3}P$. (a) Locate the kink and the marginal revenue gap. (b) Find the optimum when $MC = 20$. (c) What happens if MC falls from 9 to 5? [§16.6]
- Q77.** A monopolist can segment two markets: $Q_1 = 60 - 2P_1$ and $Q_2 = 80 - 4P_2$, with constant $MC = 10$. Find prices and quantities, compute both elasticities, and verify the inverse-elasticity rule. [§17.3]
- Q78.** Now the segments are $Q_1 = 40 - P_1$ and $Q_2 = 12 - 2P_2$ with $C = Q^2/2$. Find the optimum and explain what happens. [§17.3]
- Q79.** A firm charges 600 in one segment where the elasticity is 2.5, and 900 in another. (a) What is its marginal cost? (b) What elasticity must it be assuming in the second segment? (c) Which segment is more price sensitive? [§15.1, §17.3]
- Q80.** Rank these from best to worst for the seller, with one clause of justification each, and add the caveat that could reverse two of them: a single per-unit price; first-degree price discrimination; third-degree price discrimination; a two-part tariff. [§17.5]
- Q81.** Five hundred identical consumers each have demand $Q = 30 - P$; marginal cost is 6 and there are no fixed costs. Find the optimal two-part tariff and total profit, and compare with simple monopoly pricing. [§18.1]
- Q82.** A firm faces 200 consumers of type A with $Q = 20 - P$ and 100 of type B with $Q = 40 - P$; marginal cost is zero. Find the optimal two-part tariff. [§18.2–18.3]
- Q83.** A club must charge a usage price of 10 by regulation. It faces two types, $Q = 50 - 2P$ and $Q = 60 - P$, and wants to serve both. What membership fee should it set? [§18.2]
- Q84.** Four consumers have reservation prices (r_1, r_2) equal to $(90, 20)$, $(70, 50)$, $(50, 70)$ and $(20, 90)$; costs are zero. Compare selling separately with pure bundling, and explain the result. [§19.1–19.3]
- Q85.** A monopolist in two markets faces 5,000 consumers whose reservation values r_1 and r_2 are *independent* draws from $U[0, 60]$; costs are zero. (a) Find revenue from selling the goods separately. (b) Find the optimal pure-bundle price and revenue. (c) Explain why bundling helps here even though the valuations are independent. (d) Would mixed bundling do better, and why? [§19.4]
- Q86.** With $p_1 = 40$, $p_2 = 30$ and a bundle price of 55, classify what each of these consumers buys: $(50, 10)$, $(35, 35)$, $(60, 50)$, $(20, 20)$. [§19.2]
- Q87.** A firm's price elasticity of demand is 2 and its advertising elasticity is 0.15, with revenue of 200 crore. (a) What should it spend on advertising? (b) Its market becomes more competitive and the price elasticity rises to 5. What now, and why? [§20.1]
- Q88.** A process yields one unit each of two goods. Demands are $Q_1 = 24 - P_1$ and $Q_2 = 30 - 3P_2$, and processing is costless. Find how much to process, how much of each good to sell, and the two prices. [§20.2]
- Q89.** A firm sells a final good with inverse demand $P = 900 - Q$; conversion costs 100 per unit downstream; the upstream division's cost is $C_u(Q) = Q^2/4$. (a) With no outside market for the input, find output and the transfer price. (b) With a competitive input market at $P_e = 100$, find how much is made and used. (c) Repeat at $P_e = 250$. (d) Find the threshold price and check it. [§21.1–21.2]
- Q90.** Final demand is $P = 200 - 4Q$; an upstream monopolist has $MC = 20$ and a downstream monopolist converts one-for-one at no extra cost. Compare the integrated firm with two independent firms on price, quantity and total profit, and name two fixes. [§21.5]

Solutions

S68. (a) $MR = 150 - 6Q = 30 \Rightarrow Q = 20, P = 150 - 60 = 90, \pi = (90 - 30)(20) - 100 = 1100$.

(b) $L = \frac{90 - 30}{90} = 2/3$, so $|E| = 1/L = 1.5$. Direct check: $|E| = \left| \frac{dQ}{dP} \right| \frac{P}{Q} = \frac{1}{3} \cdot \frac{90}{20} = 1.5$ **ok**. Note $|E| > 1$, as it must be whenever $MC > 0$.

(c) Competitive output solves $150 - 3Q = 30$, giving $Q^c = 40$, so $DWL = \frac{1}{2}(90 - 30)(40 - 20) = 600$.

(d) Price and quantity are **unchanged** at $(90, 20)$, because a fixed cost does not enter MC and therefore does not move $MR = MC$. Profit falls to $1200 - 900 = 300$. (Above a fixed cost of 1200 the firm would exit in the long run.)

(e) Revenue is maximised where $MR = 0$: $Q = 25, P = 75, |E| = 1$. Revenue there is 1875 against 1800 at the profit-maximising point, but profit is lower — **revenue maximisation ignores the cost of the extra five units**.

S69. Let the common marginal cost at the optimum be m . Then $q_1 = m/4$ and $q_2 = m/2$, so $Q = \frac{3m}{4}$ and $m = \frac{4Q}{3}$ — this is the horizontally summed marginal cost curve. Setting $MR = MC$:

$$120 - 2Q = \frac{4Q}{3} \Rightarrow 360 - 6Q = 4Q \Rightarrow Q = 36, \quad m = 48, \quad q_1 = 12, \quad q_2 = 24, \quad P = 84.$$

Check: $MC_1 = 4(12) = 48, MC_2 = 2(24) = 48$ and $MR = 120 - 72 = 48$ — all three equal, as the rule requires **ok**. The cheaper plant (flatter marginal cost) produces twice as much.

S70. (a) $AC = \frac{900}{Q} + 20$ falls throughout and always exceeds $MC = 20$, so **yes, a natural monopoly**.

(b) $MR = 120 - 2Q = 20 \Rightarrow Q = 50, P = 70, \pi = (70 - 20)(50) - 900 = 1600$.

(c) $P = MC = 20$ gives $Q = 100$; revenue 2000 against cost $900 + 2000 = 2900$, a loss of **900** — exactly the fixed cost. Efficient but **not viable without a subsidy**.

(d) $P = AC$: $120 - Q = \frac{900}{Q} + 20$, i.e. $Q^2 - 100Q + 900 = 0$, so $Q = 10$ or $Q = 90$. Take the larger root (more output, more surplus): $Q = 90$ and $P = 30$, with $AC = 900/90 + 20 = 30$ **ok**. Profit is zero and the deadweight loss is small but positive.

S71. (i) **False** — the tangency of demand to AC occurs on the *falling* part of AC , so output is below minimum-cost scale: that is excess capacity. (ii) **False** — deadweight loss comes from $P > MC$, zero profit from $P = AC$, and both hold at the tangency. (iii) **True** — price equals average cost at a point to the left of both minima.

S72. (a) Firm 1 maximises $[a - b(q_1 + q_2) - c_1]q_1$, giving the reaction function $q_1 = \frac{a - c_1 - bq_2}{2b}$; substituting firm 2's symmetric reaction and collecting terms yields $q_i = \frac{a - 2c_i + c_j}{3b}$ (§9, D13). With $a = 120, b = 1$:

$$q_1 = \frac{120 - 40 + 50}{3} = 43.33, \quad q_2 = \frac{120 - 100 + 20}{3} = 13.33,$$

so $Q = 56.67$ and $P = 63.33$. Profits: $(63.33 - 20)(43.33) = 1878$ and $(63.33 - 50)(13.33) = 178$.

(b) With $c_2 = 90$ the formula gives $q_2 = \frac{120 - 180 + 20}{3} < 0$, which is infeasible, so the true equilibrium is a **corner**: firm 2 produces nothing and firm 1 is an unconstrained monopolist, $q_1 = \frac{120 - 20}{2} = 50, P = 70$. **You must then verify that firm 2 really stays out**: its marginal cost of 90 exceeds the price of 70, so it does **ok**. Never report a negative quantity — re-solve with the firm excluded and check.

S73. The follower's reaction is $q_2 = \frac{120 - 20 - q_1}{2}$. Substituting into the leader's profit gives $\pi_1 = \left[\frac{100}{2} - \frac{q_1}{2} \right] q_1$, maximised at

$$q_1 = 50 \text{ (the monopoly quantity)}, \quad q_2 = 25, \quad Q = 75, \quad P = 45.$$

Profits: leader $(45 - 20)(50) = 1250$, follower $(45 - 20)(25) = 625$. Cournot would give each $\frac{(100)^2}{9} = 1111$, so **the leader gains and the follower loses**.

No, the leader's quantity is not a best response to the follower's: against $q_2 = 25$ her best response would be $(100 - 25)/2 = 37.5$, not 50. That is precisely the point — she deliberately over-produces, and it is credible only because the choice is made *first and irreversibly*. If she could revise after seeing q_2 , the commitment would evaporate and the outcome would be Cournot.

S74. (a) $\pi_1 = (p_1 - 5)(50 - 3p_1 + 2p_2)$; the first-order condition $50 - 6p_1 + 2p_2 + 15 = 0$ gives $p_1 = \frac{65 + 2p_2}{6}$. Symmetry: $6p = 65 + 2p$, so $p = 16.25, q = 50 - 48.75 + 32.5 = 33.75$, profit $= 11.25(33.75) = 379.7$ each.

(b) The follower's reaction is $p_2 = \frac{65 + 2p_1}{6}$. Substitute it into the leader's own demand *before* she chooses — that substitution is the whole content of moving first:

$$q_1 = 50 - 3p_1 + 2 \cdot \frac{65 + 2p_1}{6} = \frac{150 - 9p_1 + 65 + 2p_1}{3} = \frac{215 - 7p_1}{3}.$$

So the leader maximises $(p_1 - 5)\frac{215 - 7p_1}{3}$; the first-order condition $215 - 7p_1 - 7(p_1 - 5) = 250 - 14p_1 = 0$ gives

$$p_1 = \underline{17.86}, \quad p_2 = \underline{16.79}, \quad \pi_1 = \underline{385.7}, \quad \pi_2 = \underline{416.7}.$$

(c) **Both firms gain** relative to 379.7, and **the follower gains more**. That is the strategic-complements pattern: committing to a high price invites the rival to raise his too, which helps the leader, but the follower can sit just below and take share. **Careful:** this does *not* mean moving first is bad — the leader is better off than under simultaneous pricing. Only the *comparison between the two firms* favours the follower.

S75. Residual demand: $q = (200 - 2P) - (3P - 30) = 230 - 5P$, so $P = 46 - \frac{q}{5}$ and residual $MR = 46 - \frac{2q}{5}$. Setting it equal to $MC = 10$:

$$46 - \frac{2q}{5} = 10 \Rightarrow q = \underline{90}, \quad P = 46 - 18 = \underline{28}.$$

The fringe supplies $3(28) - 30 = \underline{54}$. Check: total $90 + 54 = 144 = 200 - 2(28)$ **ok**, and $P = 28 \geq 10$, so the fringe is indeed active.

S76. (a) At $P = 30$ the upper branch gives $Q = 50 - 30 = 20$ and the lower gives $30 - 10 = 20$, so the kink is at $(Q, P) = (20, 30)$. Upper branch: $P = 50 - Q$, so $MR = 50 - 2Q$, which at $Q = 20$ is $\underline{10}$. Lower branch: $P = 90 - 3Q$, so $MR = 90 - 6Q$, which at $Q = 20$ is $\underline{-30}$. The gap is $[-30, 10]$.

(b) $MC = 20$ lies *above* the gap, so the firm operates on the upper (elastic) branch: $50 - 2Q = 20$ gives $Q = \underline{15}$ and $P = \underline{35}$. Sanity check: $Q = 15 < 20$, consistent with having moved *up* the demand curve.

(c) Both 9 and 5 lie *inside* the gap, so the optimum stays at the kink in both cases: $Q = 20$, $P = 30$, and the price change is **zero**. That is the price-rigidity result — marginal cost must move outside the gap before the price moves at all.

S77. Invert: $P_1 = 30 - \frac{Q_1}{2}$ so $MR_1 = 30 - Q_1 = 10$, giving $Q_1 = \underline{20}$ and $P_1 = \underline{20}$. And $P_2 = 20 - \frac{Q_2}{4}$ so $MR_2 = 20 - \frac{Q_2}{2} = 10$, giving $Q_2 = \underline{20}$ and $P_2 = \underline{15}$. Marginal cost is constant, so the segments decouple.

Elasticities: $|E_1| = 2 \cdot \frac{20}{20} = \underline{2}$ and $|E_2| = 4 \cdot \frac{15}{20} = \underline{3}$. Rule check: $\frac{P_1}{P_2} = \frac{20}{15} = \frac{4}{3}$ and $\frac{1 - 1/|E_2|}{1 - 1/|E_1|} = \frac{2/3}{1/2} = \frac{4}{3}$ **ok**. The **less elastic segment pays more**, as the rule requires.

S78. $MR_1 = 40 - 2Q_1$; $P_2 = 6 - \frac{Q_2}{2}$ so $MR_2 = 6 - Q_2$; and $MC = Q$. Assuming both segments are served, $Q_1 = \frac{40-Q}{2}$ and $Q_2 = 6 - Q$; adding, $Q = \frac{40-Q}{2} + 6 - Q$, so $5Q = 52$ and $Q = 10.4$ — but then $Q_2 = 6 - 10.4 = \underline{-4.4} < 0$, which is impossible.

So re-solve with segment 2 excluded: $40 - 2Q_1 = Q_1$ gives $Q_1 = \underline{13.33}$, $P_1 = \underline{26.67}$ and $MC = 13.33$. Check the exclusion is right: segment 2's choke price is 6, far below the marginal cost of 13.33, so its very best customer is not worth serving **ok**. A segment is served only if its choke price exceeds marginal cost at the optimum, and a negative formula quantity is the algebraic signal that it does not.

S79. The rule, derived. $MR = P + Q \frac{dP}{dQ} = P(1 - 1/|E|)$, and setting $MR = MC$ gives $\frac{P-MC}{P} = \frac{1}{|E|}$, i.e. $P = \frac{MC}{1-1/|E|}$ (§9, D15).

(a) $600 = \frac{MC}{1 - 1/2.5} = \frac{MC}{0.6}$, so $MC = \underline{360}$.

(b) $900 = \frac{360}{1 - 1/|E|}$, so $1 - 1/|E| = 0.4$ and $|E| = \underline{1.67}$.

(c) The **first segment**, with elasticity 2.5, is more price sensitive — which is exactly why it pays the lower price. State that link explicitly; it is the inverse-elasticity rule in words.

S80.

- First-degree price discrimination** — charges every unit at its buyer's maximum willingness to pay, so the seller takes the entire consumer surplus. It is the upper bound on what any scheme can earn.
- Two-part tariff** — captures *all* the surplus when consumers are identical (set the per-unit price at marginal cost and the fee at the whole consumer surplus), but only part of it when types differ, because one fee must satisfy the weakest customer served.
- Third-degree price discrimination** — one price per observable group, so prices track willingness to pay only loosely and surplus is left within each group.
- A single per-unit price** — least responsive of all, so it leaves the most with consumers.

Caveat: the ranking of (2) and (3) can reverse when consumers are very heterogeneous, since a single two-part tariff suffers from dispersion while third-degree discrimination feeds on differences between groups. What never reverses is that (1) is first and that (2) weakly beats (4), since setting the fee to zero reproduces single pricing.

S81. Consumers are identical, so set the usage price at marginal cost and take the whole surplus as the fee:

$$P^* = \underline{6}, \quad q = \underline{24}, \quad T^* = \frac{1}{2}(30 - 6)(24) = \underline{288},$$

and total profit = $500(288) = \underline{144,000}$, with **no deadweight loss**. Simple monopoly pricing: $30 - 2Q = 6$ gives $Q = 12$, $P = 18$ and profit per consumer $(18 - 6)(12) = 144$, so 72,000 in total — **exactly half**. The intuition: the two-part tariff first makes the pie as large as possible (by pricing at marginal cost) and then eats all of it.

S82. *Serve both types.* The fee is limited by the weaker type A, $CS_A(p) = \frac{1}{2}(20 - p)^2$:

$$\pi = 300 \cdot \frac{1}{2}(20 - p)^2 + p[200(20 - p) + 100(40 - p)] = 150(20 - p)^2 + p(8000 - 300p).$$

The derivative $-300(20 - p) + 8000 - 600p = 2000 - 300p$ vanishes at $p = 6.67$, giving $T = \frac{1}{2}(13.33)^2 = 88.9$ and profit 66,667.

Serve only type B. Now $\pi = 100 \cdot \frac{1}{2}(40 - p)^2 + 100p(40 - p)$, whose derivative is $-100p \leq 0$, so $p^* = 0$, $T = \frac{1}{2}(40)^2 = 800$ and profit = 80,000.

Since $80,000 > 66,667$, the firm **excludes the 200 type-A consumers** and charges the 100 type-B consumers a fee of 800 with free usage. The lesson: **serving everybody is not always optimal** — keeping the small customers forces the fee down by more than their usage revenue is worth.

S83. At $P = 10$: *Type 1* ($Q = 50 - 2P$, choke price 25): quantity = 30, surplus = $\frac{1}{2}(25 - 10)(30) = 225$. *Type 2* ($Q = 60 - P$, choke price 60): quantity = 50, surplus = $\frac{1}{2}(60 - 10)(50) = 1250$. To keep *both* types the fee cannot exceed the smaller surplus, so $T = 225$. Note what the question removed: with the usage price fixed and both types to be served, the only decision left is the minimum-surplus step. Type 2 keeps a surplus of $1250 - 225 = 1025$, which is the information rent the firm pays for pooling.

S84. *Separately.* For good 1 the candidate prices are the reservation values: 90 sells 1 unit (90), 70 sells 2 (140), 50 sells 3 (150), 20 sells 4 (80). Best is **150**; by symmetry good 2 also gives 150, so the total is **300**. *Pure bundling.* The bundle values are 110, 120, 120, 110. Pricing at 110 sells to all four for **440**; pricing at 120 sells to two for 240. Best is **440**.

Bundling wins by 47% because the valuations are **strongly negatively correlated**: consumers who love good 1 are indifferent to good 2 and vice versa, so the *sums* (110–120) are far less dispersed than the components (20–90). A single bundle price can therefore capture nearly all of the total willingness to pay, whereas any single price for one good must either exclude the low-value consumers or leave a great deal with the high-value ones.

S85. (a) A consumer buys good 1 iff $r_1 \geq P_1$, with probability $1 - P_1/60$, so revenue per consumer is $P_1(1 - P_1/60)$, maximised at $P_1 = 30$ with revenue 15. Same for good 2, so 30 per consumer and **150,000** in total.

(b) A consumer buys iff $r_1 + r_2 \geq P$. Drawing the 60×60 square, the non-buying region is a triangle, so for $P \leq 60$,

$$\Pr(r_1 + r_2 \geq P) = 1 - \frac{1}{2}\left(\frac{P}{60}\right)^2, \quad R(P) = P - \frac{P^3}{2(60)^2}.$$

$R'(P) = 1 - \frac{3P^2}{2(3600)} = 0$ gives $P^* = 60\sqrt{2/3} = 48.99$ and $R = 32.66$ per consumer, i.e. **163,299** in total — about 9% more than separate selling. (The branch $P \geq 60$ peaks at $P = 60$ with only 150,000, so it is dominated.)

(c) Bundling does *not* require negative correlation; it requires **lower dispersion of the sum**. The sum of two independent uniforms is triangular — heavily concentrated near 60 — so a single price near 49 captures almost the whole market, whereas a single price for one good must trade off margin against coverage on a flat uniform distribution.

(d) **Yes.** Pure bundling forces a consumer who values one good highly and the other barely either to buy both or to buy nothing. Mixed bundling offers each good separately at a price p alongside the bundle at P_B , peeling off those consumers and collecting revenue the bundle would have lost, while the incremental price $P_B - p$ keeps the middle of the distribution buying the package. The gain is real but modest here — of the order of 1% — because with independent uniforms the bundle already does most of the work.

S86. Compute the four surpluses for each consumer: nothing (0), good 1 ($r_1 - 40$), good 2 ($r_2 - 30$), bundle ($r_1 + r_2 - 55$).

(r_1, r_2)	good 1	good 2	bundle	buys
(50, 10)	10	−20	5	good 1 only
(35, 35)	−5	5	15	the bundle
(60, 50)	20	20	55	the bundle
(20, 20)	−20	−10	−15	nothing

Two lessons: the first consumer's *total* value (60) exceeds the bundle price, yet she does not buy the bundle — a single good gives more surplus. The second buys the bundle although *neither* good is worth its stand-alone price.

S87. (a) $\frac{A}{PQ} = \frac{E_A}{|E_P|} = \frac{0.15}{2} = 0.075$, so $A = 0.075(200) = 15$ crore.

(b) Now $\frac{A}{PQ} = \frac{0.15}{5} = 0.03$, so $A = 6$ crore at unchanged revenue. **More elastic demand means a lower markup**, so each extra unit sold earns less contribution margin and advertising is worth less at the margin. Note that the ratio falls with price elasticity and rises with advertising elasticity — getting that direction the wrong way round is the standard error.

S88. Inverse demands: $P_1 = 24 - Q$ so $MR_1 = 24 - 2Q$ (zero at $Q = 12$); $P_2 = 10 - \frac{Q}{3}$ so $MR_2 = 10 - \frac{2Q}{3}$ (zero at $Q = 15$). Since $MC = 0$: *First branch* ($Q \leq 12$): $MR_{\text{total}} = 34 - \frac{8Q}{3} = 0$ gives $Q = 12.75 > 12$, so the first branch

does not contain the solution — good 1's marginal revenue has already turned negative. *Second branch* ($Q > 12$): good 1 contributes zero, so $MR_{\text{total}} = 10 - \frac{2Q}{3} = 0$ gives $Q = 15$. So process **15** units, sell all **15** units of good 2 at $P_2 = 10 - 5 = 5$, and sell only **12** units of good 1 at $P_1 = 24 - 12 = 12$, **disposing of 3 units**. The key step is the truncation: beyond $Q = 12$ good 1 contributes *zero*, not a negative amount, because the firm would rather destroy the unit than sell it.

S89. Here $NMR = (900 - 2Q) - 100 = 800 - 2Q$ and $MC_u = Q/2$.

(a) $800 - 2Q = \frac{Q}{2}$ gives $Q = 320$; the transfer price is $MC_u(320) = 160$ and the final price is $900 - 320 = 580$.

(b) At $P_e = 100$ the divisions decouple: upstream makes Q_1 where $\frac{Q_1}{2} = 100$, so $Q_1 = 200$; downstream sells Q_c where $800 - 2Q_c = 100$, so $Q_c = 350$. The firm **buys 150** units on the market.

(c) At $P_e = 250$: $Q_1 = 500$ and $Q_c = 275$, so the firm **sells 225** units.

(d) The net position is zero when $Q_1 = Q_c$: $2P = \frac{800 - P}{2}$, so $4P = 800 - P$ and $P = 160$ — **exactly the Case-1 transfer price**, which is the built-in consistency check. Above 160 the firm is a net seller, below it a net buyer.

S90. Integrated. $MR = 200 - 8Q = 20$ gives $Q = 22.5$, $P = 200 - 90 = 110$ and $\pi = (110 - 20)(22.5) = 2025$. *Separate.*

The downstream firm facing a wholesale price w sets $200 - 8Q = w$, i.e. $Q = \frac{200 - w}{8}$ — that is the upstream firm's

demand curve. Upstream maximises $(w - 20)\frac{200 - w}{8}$; the first-order condition $220 - 2w = 0$ gives $w = 110$, then $Q = 11.25$ and $P = 200 - 45 = 155$. Profits: upstream $90(11.25) = 1012.5$, downstream $(155 - 110)(11.25) = 506.25$, total **1518.75**.

So output *halves*, the price rises by 45, and the two firms together earn 25% less than the integrated firm: **double marginalisation** — two successive markups applied to the same unit. Consumers and both firms lose, which is why it is such a strong argument for integration. **Two fixes:** vertical integration with an internal transfer price equal to marginal cost; or a **two-part wholesale tariff** (a franchise fee plus a per-unit price equal to marginal cost), which restores the efficient quantity while letting the upstream firm capture the profit. Resale price maintenance is a third.

8 Part H — Market failure, information and the verbal questions

Questions

Q91. Demand is $P = 200 - 2Q$ and private supply is $P = 20 + Q$. Each unit produced imposes an external cost of 15 on third parties. (a) Find the market and the efficient quantities. (b) Find the optimal Pigouvian tax and verify it works. (c) Compute the deadweight loss of the unregulated market. (d) What if the external cost were given as " $C = Q$ " instead? [§22.1]

Q92. Two firms must abate 8 units between them. Firm 1's marginal abatement cost is $3a_1$ and firm 2's is a_2 . (a) Find the least-cost split and its total cost. (b) What permit price supports it, and who buys? (c) How much more does a uniform standard of 4 units each cost? [§22.2]

Q93. In the class permit game each player draws 5 independent marginal abatement costs from $U[0, 100]$, one per unit of current emissions, and receives 2 free permits. (a) Derive the equilibrium permit price. (b) What is the expected net trade of a single player, and why is that consistent with active trading? [§22.3]

Q94. A factory and a laundry are neighbours. Payoffs (factory, laundry) are: (Filter, Clean air technology) = (300, 300); (Filter, No technology) = (300, 500); (No filter, Technology) = (500, 200); (No filter, No technology) = (500, 100). (a) Find the Nash equilibrium and the efficient outcome. (b) Find the bargained outcome when the factory has the right to pollute, and when the laundry has the right to clean air. (c) State the Coase theorem and three conditions under which it fails. [§22.4]

Q95. Three residents value street lighting with $MB_1 = 30 - Q$, $MB_2 = 20 - 2Q$ and $MB_3 = 12 - Q$; the marginal cost is 25. (a) Find the efficient quantity and the Lindahl prices. (b) How much would resident 1 provide alone? (c) Why does the answer to (b) understate the problem in a large population? [§23.2–23.3]

Q96. Classify each and name its characteristic market failure: a congested public road; an uncongested toll bridge; national defence; the Arctic cod fishery; a patented drug formula. [§23.1]

Q97. A clinic owner cannot observe her doctor's effort, only revenue, which is 200 or 0. High effort costs 40 and succeeds with probability 0.9; medium costs 15 and succeeds with probability 0.6; low costs 0 and succeeds with probability 0.2. Both are risk neutral, the doctor's outside option is 0, and wages cannot be negative. (a) What wage spread induces high effort? (b) What is the owner's maximum expected profit? (c) Could she instead induce medium effort, and would she want to? [§24.4]

Q98. Classify each as primarily adverse selection or moral hazard, with a one-sentence justification: (a) a university requires 100% attendance and tracks it; (b) dating sites ban photos from your picture library and demand current ones; (c) patients worry that doctors order unnecessary tests; (d) graduates from strong colleges name the college on their CV while others name only the university; (e) firms hire consultants to supervise their use of AI tools. [§24.3]

Q99. Explain in about 90 words why basic scientific research funded by universities is under-provided by the private market, being specific about who bears the cost and who receives the benefit. [§25.2]

Q100. Street vendors sell pani puri only by the plate at a fixed price and refuse to sell single pieces, although splitting the portion is easy. (a) Give the economic explanation in about 100 words. (b) Write a prompt that would get a language model to produce that explanation without telling it the answer. [S25.5]

Solutions

S91. (a) Market: $200 - 2Q = 20 + Q$ gives $Q^m = 60$ and $P = 80$. Efficient: $MSC = 20 + Q + 15 = 35 + Q$, so $200 - 2Q = 35 + Q$ gives $Q^* = 55$, with consumers paying 90.

(b) The tax equals the marginal external cost at the efficient quantity, here the constant 15. Check: supply becomes $P = 35 + Q$, which meets demand at $Q = 55$ **ok**.

(c) $DWL = \frac{1}{2}(15)(60 - 55) = 37.5$ — the triangle between MSC and demand over the excess 5 units.

(d) If $C = Q$ is the *marginal* external cost, then $MSB = 200 - 2Q - Q$ and the efficient quantity solves $200 - 3Q = 20 + Q$, giving $Q^* = 45$ and a tax of $MEC(45) = 45$. If it is the *total* external cost, the marginal external cost is the constant 1 and $Q^* = 59.67$. **Always state in one clause which reading you are using** — his 2018 paper used the marginal reading, and the two answers differ enormously.

S92. (a) Efficiency requires equal marginal abatement costs: $3a_1 = a_2$ with $a_1 + a_2 = 8$, so $4a_1 = 8$ and

$$a_1 = 2, \quad a_2 = 6.$$

$$\text{Total cost} = \int_0^2 3a \, da + \int_0^6 a \, da = 6 + 18 = 24.$$

(b) The common marginal cost is $3(2) = 6 = 1(6)$, so the permit price is 6. Firm 2, whose abatement is cheap, abates a lot and **sells** permits; firm 1 abates little and **buys** them.

(c) A uniform standard of 4 each costs $\int_0^4 3a \, da + \int_0^4 a \, da = 24 + 8 = 32$, which is **8 more** — a third higher for exactly the same environmental outcome. The reason is that a uniform standard forces the high-cost abater to do as much as the low-cost one, violating the equal-marginal-cost condition.

S93. (a) At a price P a unit is abated if and only if its marginal abatement cost is below P , which happens with probability $P/100$. Expected abatement per player is $5P/100$, and the required abatement is $5 - 2 = 3$ units. Market clearing:

$$5 \cdot \frac{P^*}{100} = 3 \Rightarrow P^* = 100 \times \frac{5-2}{5} = 60.$$

(b) At that price expected abatement is 3 units, so expected emissions are $5 - 3 = 2$, exactly the number of permits held: the **expected net trade is zero**. That is consistent with heavy trading because it is only the *average* that nets out — a player who happens to draw five low costs abates everything and sells two permits, while one who draws five high costs abates nothing and buys three. **Trade reallocates permits from low-cost to high-cost abaters**, which is exactly what makes the outcome cost-minimising.

S94. (a) “No filter” pays the factory 500 against 300 whatever the laundry does, so it is **strictly dominant**; given that, the laundry compares 200 with 100 and adopts the technology. The Nash equilibrium is **(No filter, Technology)** with a total of 700. Joint surpluses are 600, **800**, 700 and 600, so the efficient profile is **(Filter, No technology)**: it is cheaper to stop the pollution at source than to have the laundry defend against it.

(b) *Factory holds the right*. The disagreement point is the Nash outcome (500, 200). Moving to the efficient profile creates 800, a gain of 100; splitting it equally, the laundry pays the factory 250 to install the filter, giving (550, 250). *Laundry holds the right*. The factory must filter, and the outcome is (300, 500) directly — the factory would have to compensate the laundry 400 to gain only 200, so it does not try.

(c) **Coase theorem**: with clearly assigned property rights and costless bargaining, the parties reach the efficient outcome *regardless of who holds the right*; the assignment determines only the distribution of surplus. It fails with **transaction costs**; with **many parties** (free riding and hold-out); with **asymmetric information** about costs, so that efficient deals fall through; and with ill-defined or unenforceable rights.

S95. (a) A public good is non-rival, so add marginal benefits *vertically*. Resident 2 drops out at $Q = 10$ and resident 3 at $Q = 12$, so for $Q \leq 10$,

$$MSB = (30 - Q) + (20 - 2Q) + (12 - Q) = 62 - 4Q = 25 \Rightarrow Q^* = 9.25,$$

which is below 10, so the branch is valid. Lindahl prices are each resident’s own marginal benefit at Q^* :

$$30 - 9.25 = 20.75, \quad 20 - 18.5 = 1.5, \quad 12 - 9.25 = 2.75,$$

summing to 25 **ok** — the personalised prices exactly finance the marginal cost.

(b) Acting alone, resident 1 buys where her *own* marginal benefit equals the cost: $30 - Q = 25$, so $Q = 5$, a little over half the efficient amount, with the other two free-riding entirely.

(c) With three residents one high-value person still provides a substantial amount. In a large population **each individual’s marginal benefit is a tiny fraction of the social total**, so the largest single beneficiary provides almost nothing relative to Q^* , and the shortfall approaches 100%. That is why public goods are tax-financed rather than left to the most enthusiastic user.

S96.

	Classification	Characteristic failure
Congested public road	common resource (rival through congestion, non-excludable)	over-use
Uncongested toll bridge	club good (non-rival, excludable)	inefficient exclusion — the toll deters users whose crossing costs nothing
National defence	public good	under-provision through free riding
Arctic cod fishery	common resource	over-fishing, the tragedy of the commons
Patented drug formula	club good made from a public good by legal exclusion	static loss (price above zero marginal cost) traded for dynamic gain (the incentive to invent)

The distinction that matters: a *common resource* fails through **over-use**, a *public good* through **under-provision** — different failures needing different remedies.

S97. Set the wage at zero when revenue is zero, so only the spread $\Delta = w_{200} - w_0$ provides incentives. The doctor's expected utilities are

$$\text{high} = 0.9\Delta - 40, \quad \text{medium} = 0.6\Delta - 15, \quad \text{low} = 0.2\Delta.$$

(a) High beats medium iff $0.3\Delta \geq 25$, i.e. $\Delta \geq 83.33$; high beats low iff $0.7\Delta \geq 40$, i.e. $\Delta \geq 57.14$. The **binding** constraint is the first, so $\Delta = 83.33$ and **the next-best option is medium effort**, not low. (Contrast the 2018 clinic question, where low effort was the binding alternative — you must check both.)

(b) With $w_0 = 0$ and $w_{200} = 83.33$: expected revenue $= 0.9(200) = 180$, expected wage $= 0.9(83.33) = 75$, so profit $= 105$. Participation holds, since the doctor earns $75 - 40 = 35 > 0$ — his **information rent**, which limited liability forces the owner to pay.

(c) **Yes, medium effort can be induced**: it needs $\Delta \geq \frac{15}{0.4} = 37.5$ to beat low and $\Delta \leq 83.33$ to beat high, so $\Delta = 37.5$ works. Profit would be $0.6(200) - 0.6(37.5) = 97.5$. Paying nothing and accepting low effort gives $0.2(200) = 40$. So the owner **prefers high effort** (105), but note how close medium is — if high effort cost a little more, the ranking would flip.

S98.

	Diagnosis	Why
(a) Attendance tracking	Moral hazard	The university has already selected its students; the unobserved thing is their <i>effort</i> after admission, and monitoring restores observability.
(b) Current photos only	Adverse selection	The hidden thing is a <i>characteristic</i> that exists before matching; forcing current photos stops misrepresentation of type.
(c) Unnecessary tests	Moral hazard	The doctor's <i>action</i> is unobservable to the patient and her fee is tied to procedures rather than outcomes.
(d) Naming the college	Adverse selection	Graduates of strong colleges signal their hidden type; graduates of weak ones suppress the signal, so the disclosure is informative.
(e) Consultants supervising AI	Moral hazard	A tool bears no accountability; a human agent whose reputation and fees are at stake has incentives aligned with the client's.

The one-line test: **hidden type before contracting is adverse selection; hidden action after contracting is moral hazard.**

S99. Model answer. “Basic research is close to a pure public good: once a result is published it is non-rival — any number of firms can use it at no extra cost — and largely non-excludable, since patents cover applications rather than underlying knowledge. A private funder therefore compares its own appropriable return with the full cost, ignoring the spillovers to rival firms, to future innovators and to patients. Because the social return vastly exceeds the private return, the private market invests too little: the cost is borne by one funder while the benefit is dispersed across the whole economy. Long horizons and high failure rates compound the gap, which is the standard case for public funding.”

S100. (a) Model answer. “The binding constraint for the vendor is not ingredients but *service time*: the marginal cost of one more piece is negligible, while the marginal cost of one more customer interaction is large, because capacity is the number of customers he can serve in an hour. A fixed plate is therefore a minimum-quantity requirement — effectively a two-part tariff collapsed into a single price — which captures much more of each buyer's willingness to pay than per-piece pricing and prevents low-value buyers from occupying a full service slot for one piece. It also standardises the product, making price and quality easy to compare, and simplifies queue management. He is maximising profit per unit of serving time, not per piece.”

(b) A prompt should supply a *role*, the *economic frame* to use, the real *constraints*, the required *form*, and a *quality bar* — while leaving the conclusion to be derived. For example:

“You are a microeconomist explaining everyday pricing decisions to a first-year MBA class. A street food seller offers only a fixed-size plate at a single price and refuses to sell smaller quantities, even though splitting the portion is physically trivial. Explain the demand-side and cost-side reasons a seller would design the offer this way rather than pricing per piece. Cover at least: how per-transaction and per-customer costs compare with the cost of the food itself; how a fixed portion affects the seller's ability to capture willingness to pay relative to linear pricing; what queueing and service time imply when capacity, not ingredients, is the binding constraint; and what happens to the seller's surplus if buyers may choose their own quantity. State explicitly what the seller is maximising. Avoid jargon beyond standard terms, include one concrete numerical illustration, and keep it under 400 words.”

Notice what the prompt does *not* do: it never says “two-part tariff”, “minimum quantity” or “profit per unit of time”. It names the *dimensions* to reason about and lets the model supply the mechanism — which is what makes it a fair prompt rather than a restatement of the answer.

9 Derivations: every standard formula built from scratch

Elsewhere in these notes the standard results are stated as formulas you can apply under time pressure. This section derives every one of them. Use it two ways: read it once now, so that no formula in the notes is a black box; and keep it to hand in the exam, because a descriptive paper can ask you to *derive* rather than to *apply*, and a question that says “show that” or “derive the equilibrium” is asking for exactly the steps below. Each derivation is written the way you should write it on the answer sheet: state the maximisation, take the first-order condition, impose equilibrium, solve.

9.1 D1. First-price sealed bid: why $b(v) = \frac{n-1}{n}v$

n bidders, private values drawn independently from $U[0, 1]$, highest bid wins and pays its own bid. Look for a symmetric equilibrium in which everyone uses the same strictly increasing bid function $b(\cdot)$.

The trick is to let a bidder of true value v consider *pretending* to be some other type v' — that is, submitting $b(v')$ — and then to impose that truth-telling is optimal. Because b is increasing, bidding $b(v')$ wins exactly when every one of the $n - 1$ rivals has a value below v' , which happens with probability $(v')^{n-1}$. So the bidder solves

$$\max_{v'} (v')^{n-1} (v - b(v')).$$

Differentiate with respect to v' and then set $v' = v$, which is what “truthful bidding is optimal” means:

$$(n - 1)v^{n-2}(v - b(v)) - v^{n-1}b'(v) = 0.$$

Multiply through by v and rearrange into a form you can integrate:

$$v^{n-1}b'(v) + (n - 1)v^{n-2}b(v) = (n - 1)v^{n-1} \Rightarrow \frac{d}{dv} [v^{n-1}b(v)] = (n - 1)v^{n-1}.$$

The left side is an exact derivative — that is the step worth spotting. Integrate from 0 to v , using $b(0) = 0$ (a bidder with the lowest possible value never bids a positive amount):

$$v^{n-1}b(v) = \frac{n-1}{n}v^n \Rightarrow \boxed{b(v) = \frac{n-1}{n}v}$$

Reading it. $\frac{n-1}{n}v$ is precisely $\mathbb{E}[\text{highest rival value} \mid \text{your value } v \text{ is the highest}]$, so the equilibrium bid is “what I expect the competition to be”. Shading $\frac{1}{n}$ of your value shrinks as n grows, because the cost of losing rises.

9.2 D2. Order statistics for n draws from $U[0, 1]$

These two numbers are needed for every revenue comparison, so derive them once.

Highest of n .

$\Pr(\max \leq x) = x^n$, so the density is nx^{n-1} and

$$\mathbb{E}[\max] = \int_0^1 x \cdot nx^{n-1} dx = \frac{n}{n+1}.$$

Second highest of n .

The second highest lies in $[x, x + dx]$ when one of the n draws is in that interval (n ways), exactly one of the remaining $n - 1$ is above (that is $(n - 1)(1 - x)$ ways), and the other $n - 2$ are below (probability x^{n-2}). So the density is $n(n - 1)x^{n-2}(1 - x)$ and

$$\mathbb{E}[2\text{nd}] = \int_0^1 x \cdot n(n - 1)x^{n-2}(1 - x) dx = n(n - 1) \left[\frac{1}{n} - \frac{1}{n+1} \right] = \frac{n-1}{n+1}.$$

Sanity check: $n = 2$ gives $\mathbb{E}[\max] = 2/3$ and $\mathbb{E}[2\text{nd}] = 1/3$, which sum to 1 — correct, since the two values together are just two uniform draws averaging $1/2$ each.

9.3 D3. Revenue equivalence, verified in both formats

Second price.

Bidding your value is weakly dominant (D6), so the winner pays the second-highest value and expected revenue is

$$\mathbb{E}[2\text{nd}] = \frac{n-1}{n+1}.$$

First price.

The winner is the highest-value bidder and pays $\frac{n-1}{n}$ of her value, so expected revenue is

$$\frac{n-1}{n} \mathbb{E}[\max] = \frac{n-1}{n} \cdot \frac{n}{n+1} = \frac{n-1}{n+1}.$$

Identical — and that is revenue equivalence in a single line. The two formats move the money differently (one shades bids, the other shades the price) and the two effects cancel exactly.

What the theorem actually requires: risk-neutral bidders; independent private values from a common distribution; the object always goes to the highest bidder; and the lowest possible type earns zero expected surplus. Break any one and revenue equivalence fails — risk aversion favours first price, affiliated values favour ascending, and security bids break it in the direction derived in §8.

9.4 D4. All-pay auction: $b(v) = \frac{n-1}{n} v^n$

Same setup as D1, except that *every* bidder pays her bid whether or not she wins. Pretending to be v' now gives

$$\max_{v'} (v')^{n-1} v - b(v'),$$

since the payment no longer depends on winning. The first-order condition at $v' = v$ is simply

$$b'(v) = (n-1)v^{n-2} \cdot v = (n-1)v^{n-1} \Rightarrow b(v) = \frac{n-1}{n} v^n \quad (\text{using } b(0) = 0).$$

Expected revenue is $n \mathbb{E}[b(v)] = n \int_0^1 \frac{n-1}{n} v^n dv = \frac{n-1}{n+1}$ — the same again, as revenue equivalence promises. Note the shape: all-pay bids are convex in v , so low types bid almost nothing and high types bid close to their value.

9.5 D5. The optimal reserve price is $1/2$, whatever n is

Seller values the object at 0, n bidders with values $\sim U[0, 1]$, second-price auction with reserve r .

Expected revenue has two pieces: the reserve is paid when exactly one bidder is above it, and otherwise the second-highest value is paid.

$$\Pi(r) = \underbrace{r \cdot nr^{n-1}(1-r)}_{\text{exactly one bidder above } r} + \underbrace{\int_r^1 y n(n-1)y^{n-2}(1-y) dy}_{\text{two or more above}}.$$

Differentiating and simplifying gives $\Pi'(r) = nr^{n-1}(1-2r)$, so $\Pi'(r) = 0$ at $\boxed{r^* = 1/2}$ for every n .

The one-line argument to write instead, if you are short of time. Raising the reserve matters only in the event that the highest value is near r — in every other state the price is set by the second-highest bid and the reserve is irrelevant. At that margin the seller trades the probability of no sale against a higher price, and the comparison is between her own value (0) and r , which does not involve n . So n cannot enter the answer. What n does change is how *often* the reserve binds (probability r^n , which vanishes), so the reserve matters less and less as the market thickens.

9.6 D6. Second price: bidding your value is weakly dominant

Let $\hat{b}$ be the highest rival bid. Your bid decides only *whether* you win, never *what* you pay. Compare bidding v with bidding any $b \neq v$:

- *Bidding $b > v$.* This changes the outcome only when $v < \hat{b} < b$: you now win and pay $\hat{b} > v$, a loss. Otherwise the outcome is identical.
- *Bidding $b < v$.* This changes the outcome only when $b < \hat{b} < v$: you now lose an auction you would have won at a price below your value, forgoing a gain.

So v is at least as good as any other bid against every rival profile, and strictly better against some. That is weak dominance. Note what the argument never used: the number of bidders, the distribution of values, or your risk attitude — which is why the result is so robust, and why he tests it.

9.7 D7. Tullock contest: $x^* = \frac{(n-1)V}{n^2}$

n identical players, prize V , effort x_i costing x_i , win probability $x_i / \sum_j x_j$.

Write $X_{-i} = \sum_{j \neq i} x_j$. Player i maximises

$$\frac{x_i}{x_i + X_{-i}} V - x_i, \quad \text{first-order condition} \quad \frac{X_{-i}}{(x_i + X_{-i})^2} V = 1.$$

(The derivative of $\frac{x_i}{x_i + X_{-i}}$ is $\frac{X_{-i}}{(x_i + X_{-i})^2}$ by the quotient rule.) Now impose symmetry: $x_i = x$ for all i , so $X_{-i} = (n-1)x$ and $x_i + X_{-i} = nx$:

$$\frac{(n-1)x}{n^2 x^2} V = 1 \Rightarrow \boxed{x^* = \frac{(n-1)V}{n^2}}$$

Total spending is $nx^* = \frac{(n-1)V}{n}$, which tends to V : **full rent dissipation**. Each player's profit is $\frac{V}{n} - x^* = \frac{V}{n^2}$, which collapses faster still.

9.8 D8. Asymmetric contest: $x_i = \frac{V_i^2 V_j}{(V_i + V_j)^2}$

Two players with different prizes V_1, V_2 . The two first-order conditions are

$$\frac{x_2}{(x_1 + x_2)^2} V_1 = 1 \quad \text{and} \quad \frac{x_1}{(x_1 + x_2)^2} V_2 = 1.$$

Both left sides share the denominator $(x_1 + x_2)^2$, so dividing one by the other kills it and gives the key ratio immediately:

$$\frac{x_2 V_1}{x_1 V_2} = 1 \Rightarrow \frac{x_1}{x_2} = \frac{V_1}{V_2}.$$

So efforts are in proportion to prizes — and therefore **player i wins with probability $V_i/(V_1 + V_2)$** , which is the result worth quoting even if you never finish the algebra. To finish it, write $x_2 = x_1 V_2/V_1$, so $x_1 + x_2 = x_1(V_1 + V_2)/V_1$, and substitute into the first condition:

$$\frac{x_1 V_2/V_1}{x_1^2 (V_1 + V_2)^2/V_1^2} V_1 = 1 \Rightarrow \frac{V_1^2 V_2}{x_1 (V_1 + V_2)^2} = 1 \Rightarrow \boxed{x_1 = \frac{V_1^2 V_2}{(V_1 + V_2)^2}}$$

and symmetrically for x_2 . *Careful:* each player's spending as a share of her *own* prize is $x_i/V_i = V_i V_j/(V_i + V_j)^2$, which is the *same* for both. The underdog is deterred rather than provoked by a larger gap, and if V_2 is small enough relative to V_1 the corner solution $x_2 = 0$ takes over — always check it.

9.9 D9. Grim trigger: $\delta \geq \frac{T-R}{T-P}$

A stage game repeated forever with discount factor δ . Cooperating forever pays R each period; defecting pays T once (the temptation) and then P forever (the punishment), because the trigger is pulled and never released.

Compare the two infinite streams, using $\sum_{t \geq 0} \delta^t = \frac{1}{1-\delta}$:

$$\underbrace{\frac{R}{1-\delta}}_{\text{cooperate}} \geq \underbrace{T + \frac{\delta P}{1-\delta}}_{\text{defect once, then punished}}.$$

Multiply through by $(1-\delta) > 0$ and collect the δ terms:

$$R \geq T(1-\delta) + \delta P \Rightarrow R - T \geq \delta(P - T) \Rightarrow \delta \geq \frac{T - R}{T - P},$$

where the inequality flips at the last step because $P - T < 0$.

$$\boxed{\delta \geq \frac{T - R}{T - P}}$$

Reading it. The numerator is the *one-period* gain from cheating; the denominator is the *per-period* loss you suffer forever afterwards. Note that the sucker's payoff never appears: grim trigger never puts you in that position on the equilibrium path.

9.10 D10. Bertrand cartel with n firms: $\delta \geq 1 - \frac{1}{n}$

Apply D9 with the three numbers this game supplies. Colluding at the monopoly price gives each firm $R = \pi^m/n$. Undercutting by a hair captures the entire market for one period, so $T = \pi^m$. The punishment is Bertrand competition at marginal cost, so $P = 0$. Then

$$\delta \geq \frac{\pi^m - \pi^m/n}{\pi^m - 0} = 1 - \frac{1}{n}.$$

So two firms need $\delta \geq 0.5$, four firms $\delta \geq 0.75$, ten firms $\delta \geq 0.9$: **collusion gets harder as the cartel grows**, because your collusive share shrinks while the prize from cheating stays the whole market.

Under Cournot instead of Bertrand, the punishment payoff P is positive rather than zero, which weakens the threat — but the temptation T also falls, because a deviator can only expand output rather than seize the whole market. The second effect dominates: the linear Cournot duopoly threshold is $9/17 \approx 0.53$ against 0.5 for Bertrand. State that trade-off rather than just quoting a number.

9.11 D11. Rubinstein bargaining: the proposer gets $\frac{1-\delta_2}{1-\delta_1\delta_2}$

A pie of size 1, alternating offers, infinite horizon, discount factors δ_1, δ_2 . Player 1 proposes in odd rounds, player 2 in even rounds.

Stationarity is the whole trick: the game after any rejection looks exactly like the game at the start with the roles swapped, so define

X = player 1's share when *she* proposes, Y = player 1's share when *he* proposes.

A proposer offers the responder exactly her discounted continuation value — no more (it would be wasteful) and no less (it would be rejected). This gives two equations:

- When player 1 proposes, she must give player 2 the value $\delta_2(1 - Y)$, since player 2 would get $1 - Y$ next round. So $X = 1 - \delta_2(1 - Y)$.
- When player 2 proposes, he gives player 1 the value $\delta_1 X$. So $Y = \delta_1 X$.

Substitute the second into the first:

$$X = 1 - \delta_2 + \delta_2\delta_1 X \Rightarrow X(1 - \delta_1\delta_2) = 1 - \delta_2 \Rightarrow X = \frac{1 - \delta_2}{1 - \delta_1\delta_2}$$

Checks worth doing aloud. With $\delta_1 = \delta_2 = \delta$ this is $\frac{1-\delta}{1-\delta^2} = \frac{1}{1+\delta}$ — the familiar symmetric answer, which tends to $1/2$ as $\delta \rightarrow 1$ (patience erases the first-mover advantage) and to 1 as $\delta \rightarrow 0$ (an impatient responder accepts anything). And X is *decreasing* in δ_2 : the more patient your opponent, the less you keep.

9.12 D12. The Nash bargaining solution

Maximise the Nash product $(u_1 - d_1)(u_2 - d_2)$ subject to $u_1 + u_2 = S$. Substitute $u_2 = S - u_1$ and differentiate:

$$\frac{d}{du_1} [(u_1 - d_1)(S - u_1 - d_2)] = (S - u_1 - d_2) - (u_1 - d_1) = 0 \Rightarrow u_1 = d_1 + \frac{S - d_1 - d_2}{2}.$$

So each party gets **its own disagreement payoff plus half the surplus above disagreement**. Two consequences to quote: $\partial u_1 / \partial d_1 = \frac{1}{2}$, so *half of any improvement in your threat point leaks to the other side*; and the axioms behind the solution are Pareto efficiency, symmetry, invariance to rescaling utilities, and independence of irrelevant alternatives. The “Nash programme” link is that Rubinstein’s non-cooperative answer (D11) converges to this as $\delta \rightarrow 1$.

9.13 D13. Cournot with different costs: $q_i = \frac{a-2c_i+c_j}{3b}$

Inverse demand $P = a - b(q_1 + q_2)$, constant marginal costs c_1, c_2 . Firm 1 maximises

$$(a - b(q_1 + q_2) - c_1)q_1 \Rightarrow \text{FOC: } a - 2bq_1 - bq_2 - c_1 = 0 \Rightarrow q_1 = \frac{a - c_1 - bq_2}{2b},$$

which is firm 1's reaction function; firm 2's is symmetric. Substitute firm 2's into firm 1's:

$$q_1 = \frac{a - c_1}{2b} - \frac{1}{2} \cdot \frac{a - c_2 - bq_1}{2b} = \frac{a - c_1}{2b} - \frac{a - c_2}{4b} + \frac{q_1}{4}.$$

Collect q_1 : $\frac{3}{4}q_1 = \frac{2(a - c_1) - (a - c_2)}{4b}$, so

$$q_1 = \frac{a - 2c_1 + c_2}{3b}$$

How to use it safely. The coefficient -2 on your own cost and $+1$ on your rival's says that a cost advantage helps you roughly twice as much as it hurts your rival. If the formula returns a *negative* quantity, the interior solution is invalid: re-solve with that firm excluded, and then **verify the exclusion** by checking that its marginal cost exceeds the resulting monopoly price. Never report a negative quantity.

9.14 D14. Stackelberg: why the leader produces the monopoly quantity

With $P = a - bQ$ and common marginal cost c , the follower's reaction (from D13 with $c_1 = c_2 = c$) is $q_2 = \frac{a - c - bq_1}{2b}$. The leader substitutes it *before* choosing, which is the entire content of moving first:

$$\pi_1 = \left(a - bq_1 - b \cdot \frac{a - c - bq_1}{2b} - c \right) q_1 = \frac{(a - c) - bq_1}{2} q_1.$$

The first-order condition $\frac{a-c}{2} - bq_1 = 0$ gives

$$q_1 = \frac{a - c}{2b} \text{ (exactly the monopoly quantity), } q_2 = \frac{a - c}{4b}, \quad Q = \frac{3(a - c)}{4b}, \quad P = \frac{a + 3c}{4}.$$

The point to state. The leader's quantity is **not** a best response to the follower's: against q_2 her best reply would be smaller. She deliberately over-produces, and this is credible only because the choice is made first and irreversibly. If she could revise after observing q_2 , the commitment would evaporate and the outcome would collapse back to Cournot. *Commitment works by destroying your own option to be flexible.*

9.15 D15. The Lerner index and the inverse-elasticity rule

Revenue is PQ , so marginal revenue is

$$MR = \frac{d(PQ)}{dQ} = P + Q \frac{dP}{dQ} = P \left(1 + \frac{Q}{P} \frac{dP}{dQ} \right) = P \left(1 - \frac{1}{|E|} \right),$$

using $|E| = -\frac{dQ}{dP} \frac{P}{Q}$. Setting $MR = MC$ and rearranging,

$$P \left(1 - \frac{1}{|E|} \right) = MC \Rightarrow \boxed{\frac{P - MC}{P} = \frac{1}{|E|}} \quad \text{equivalently} \quad P = \frac{MC}{1 - 1/|E|}.$$

Two immediate consequences: since $MC > 0$ forces $MR > 0$, a monopolist **always** operates where $|E| > 1$; and across two segments of a price-discriminating monopolist, $\frac{P_1}{P_2} = \frac{1 - 1/|E_2|}{1 - 1/|E_1|}$, so the **less elastic segment pays more**. *Careful:* the Lerner index is a *ratio*, not the difference $P - MC = 1/|E|$, and a higher index means a higher markup, not higher profit.

9.16 D16. Advertising: $\frac{A}{PQ} = \frac{E_A}{|E_P|}$

Let $Q(P, A)$ depend on both price and advertising spend A , with profit $\pi = (P - c)Q(P, A) - A$. Two first-order conditions:

$$\frac{\partial \pi}{\partial A} = (P - c)Q_A - 1 = 0 \Rightarrow (P - c)Q_A = 1,$$

$$\frac{\partial \pi}{\partial P} = Q + (P - c)Q_P = 0 \Rightarrow (P - c) = -\frac{Q}{Q_P} = \frac{P}{|E_P|},$$

the last step using $|E_P| = -Q_P P / Q$. Now write the advertising elasticity $E_A = \frac{A}{Q} Q_A$. From the first condition $Q_A = \frac{1}{P - c}$, so

$$E_A = \frac{A}{Q(P - c)} = \frac{A}{Q} \cdot \frac{|E_P|}{P} = \frac{A}{PQ} |E_P| \Rightarrow \boxed{\frac{A}{PQ} = \frac{E_A}{|E_P|}}$$

Advertising intensity rises with how responsive demand is to advertising and falls with how price-sensitive customers are — which is why habit goods with inelastic demand advertise heavily.

9.17 D17. Spence signalling: the separating range

Wages $w_H > w_L$, education costs c_H per unit for the high type and c_L for the low type, with $c_H < c_L$ (**single crossing**: education is cheaper for the able). For a separating equilibrium at threshold y^* , two incentive constraints must hold simultaneously:

- The low type must not mimic: $w_L \geq w_H - c_L y^* \Rightarrow y^* \geq \frac{w_H - w_L}{c_L}$.
- The high type must be willing to signal: $w_H - c_H y^* \geq w_L \Rightarrow y^* \leq \frac{w_H - w_L}{c_H}$.

$$\boxed{\frac{w_H - w_L}{c_L} \leq y^* \leq \frac{w_H - w_L}{c_H}}$$

The interval is non-empty exactly when $c_H < c_L$ — single crossing is not decoration, it is the condition that makes signalling possible at all. Note that education here is pure waste: it raises no one's productivity and exists only to separate types, which is why the separating equilibrium can be worse for everyone than pooling would be.

9.18 D18. Lemons and the takeover problem: $B^* = \frac{a}{2-k}$

The target's value v is uniform on $[a, b]$ and known only to the seller. The buyer can run it better, so it is worth kv to her. She offers B ; the seller accepts iff $v \leq B$.

The whole difficulty is the conditional expectation. Acceptance tells the buyer that $v \leq B$, so the relevant value is $\mathbb{E}[v \mid v \leq B] = \frac{a+B}{2}$ — the midpoint of $[a, B]$, *not* the unconditional mean $\frac{a+b}{2}$. Expected profit is

$$\Pi(B) = \underbrace{\frac{B-a}{b-a}}_{\text{Pr(accept)}} \left(k \cdot \frac{a+B}{2} - B \right) = \frac{1}{b-a} (B-a) \left(\frac{ka}{2} - mB \right), \quad m \equiv 1 - \frac{k}{2} = \frac{2-k}{2}.$$

Differentiate by the product rule and set to zero:

$$\frac{ka}{2} - mB - m(B-a) = 0 \Rightarrow 2mB = \frac{ka}{2} + ma = \frac{ka + (2-k)a}{2} = a,$$

because the k terms cancel exactly — that cancellation is the whole reason the answer is so clean. Hence $B = \frac{a}{2m}$, that is

$$\boxed{B^* = \frac{a}{2-k}} \quad (k < 2; \text{ the second derivative is } \frac{2-k}{a-b} < 0).$$

Reading it. The answer depends on the **lower** endpoint a and not at all on the upper one. With $a = 0$ it gives $B^* = 0$: **the market shuts down completely**, because however low you bid, the cars you attract average only half your bid, and $k < 2$ means half of k times that is less than what you paid. With $a > 0$ trade survives, because even the worst car in the market is worth something. If $k > 2$ the bracket never turns negative and the buyer bids the maximum.

Two instances he has used: $v \sim U[0, 20]$ with $k = 1.5$ gives $B^* = 0$; $v \sim U[10, 30]$ with $k = 1.5$ gives $B^* = \frac{10}{0.5} = 20$. The general lesson is that **winning is bad news**: the very fact that your offer was accepted tells you the asset is worse than average. That is the same adverse-selection logic as the winner's curse in a common-value auction — condition on the event in which you get what you asked for, never on the unconditional average.

9.19 D19. Two-part tariffs: why $P = MC$ when consumers are identical

Charge an entry fee T plus a per-unit price P . With identical consumers the firm can set T equal to the whole consumer surplus $CS(P)$, so profit per consumer is

$$\pi(P) = CS(P) + (P - MC)Q(P) = \text{total surplus generated at price } P.$$

Total surplus is maximised at $P = MC$, and the fee extracts all of it, so the firm chooses

$$\boxed{P = MC, \quad T = CS(MC)}$$

and earns the entire efficient surplus. Any $P > MC$ shrinks the pie the firm is about to swallow whole — it is strictly self-defeating.

With different types the result breaks, because one fee must be acceptable to the *weakest* customer you intend to serve. The procedure is then: list the candidate sets of customers to serve; for each, set T equal to the *lowest* served customer's surplus and maximise $\pi(P) = (\# \text{served})T(P) + (P - MC)Q_{\text{total}}(P)$ over P ; compare the maxima. Watch for **kinks**, since the identity of the lowest-surplus type can switch as P changes — check the kink price explicitly as well as the interior first-order condition.

9.20 D20. Double marginalisation, and why the fix is marginal-cost transfer pricing

An upstream division with marginal cost c sells to a downstream division at transfer price t ; downstream faces $P = a - bQ$ and adds no further cost. Downstream maximises $(a - bQ - t)Q$, giving $Q = \frac{a-t}{2b}$. Upstream then chooses t to maximise $(t - c)\frac{a-t}{2b}$, giving $t = \frac{a+c}{2}$ and

$$Q = \frac{a-c}{4b}, \quad \text{against the integrated optimum } Q^m = \frac{a-c}{2b}.$$

So output is **half** what a single firm would produce, and the two divisions together earn less than one merged firm would. The cause is that a markup is applied twice to the same unit. The fix is to set the transfer price at the upstream division's *marginal cost* at the optimal total output, which makes the downstream division internalise the true cost; equivalently, merge the divisions, or use a two-part transfer (a lump-sum plus marginal-cost units). This is exactly why the transfer-pricing rule in §21 is " $NMR = MC_u$ at the optimum" in every one of its cases.